The galaxy-bias ladder across thin redshift bins

A correlated linear-bias prior, the one-loop bias models A and B, and the reach of DESI C_ℓ^{gg}

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1 Summary

The question. Thin redshift bins measure growth only if each bin's galaxy bias is tied to its neighbors. For DESI angular band powers C_ℓ^{gg} alone on four bin layouts (Section 2), this note sets how freely a thin bin's linear bias may move within its tracer group, adds the one-loop bias models A and B, tests which choices recover the cosmology honestly, and gives the figure of merit (FoM) of those that pass against the scale cut $k_{\rm max}$.

The linear-bias prior. Each bin's bias follows its group's growth-tied trend B_g/D times a free straight line in redshift, plus a residual with a correlated Gaussian prior of width σ_g and correlation length L (Section 3). Four priors span the range: growth-tied (no line, no residual), baseline ($\sigma_g = 3\%$ for BGS and 1% elsewhere, $L = 0.2$), pessimistic (3% everywhere, $L = 0.05$) and free (one free bias per bin).

Models A and B. Both add one-loop terms to each bin's linear bias: quadratic and tidal biases on calibrations plus one offset per group, a k^2 counterterm α_0, and the stochastic terms SN_0 and $\mathrm{SN}_2\, k^2\mu^2$ (Section 4). Adjacent thin bins share an edge across which the projected SN_2 is anticorrelated, which no white noise mimics; without SN_2, model A failed before its own cut. Model B also frees the redshift-space counterterm α_2 and doubles the offset widths (Section 5). Both hold one offset and counterterm per group; model A + tilts and model B + tilts also give them a free straight line in redshift within each group, as the linear bias has.

The accuracy test. At $k_{\rm max} = 0.075\,h/\mathrm{Mpc}$ every prior fits 2000 random bias histories under model A (Section 6). The growth-tied prior fails. The baseline prior is honest only when the truth is as smooth as it assumes: under the pessimistic residuals its Ω_m error bar covers the truth in 60–67% of universes instead of 68%. The pessimistic and free priors are honest under both, and model A + tilts absorbs a linear drift of the calibrations that moves model A by more than $\sigma/3$ on the wide layouts.

The reach of the models and the result. Two scenarios pair the baseline prior with models A and B and the pessimistic prior with models A and B + tilts, fitted against $k_{\rm max}$ to truths that carry α_2, SN_2, the offsets at one prior width, or the calibration drift (Figure 2); a truth inside the fitted model's own prior is reported, one outside it is held to $\sigma/3$ (Section 5). The baseline scenario fails quickly: from $k_{\rm max} = 0.1\,h/\mathrm{Mpc}$ on, its largest tested shift is 0.53–2.8σ in both models, mostly from the calibration drift. It also buys little: at 0.2 model B gives $\mathrm{FoM}(w_0, w_a) = 8.47$ and 9.92 on the thin layouts, 1.12 and 1.18 times the 7.54 and 8.43 of model B + tilts. Model A buys little even below 0.1: with tilts it holds to 0.075 on $\Delta z = 0.02$ but only to 0.05 on $\Delta d_{\rm com} = 62$ Mpc, and its $\mathrm{FoM}(w_0, w_a)$ at 0.075, 1.01 and 0.675, is small next to those of model B + tilts at 0.2. The adopted model is therefore model B + tilts

on the pessimistic prior, at every cut. Truths at one prior width move it by up to 0.99σ on $\Delta d_{\rm com} = 62$ Mpc at 0.075 and by 0.33–1.09σ on the three thinner layouts from 0.1 on, mostly through α_2, and by at most 0.22σ on $\Delta z = 0.2$; over random bias histories its error bars stay honest, $\sqrt{\langle\chi^2\rangle/n_{\rm dof}} \leq 1.028$. It stays below LSST Y1 3x2pt in both FoMs. Appendices R and S give the limits and the open issues.

2 The forecast

The data are the DESI angular band powers C_ℓ^{gg} of thin redshift bins, computed without the Limber approximation, for six tracer groups (BGS, LRG1, LRG2, LRG3, ELG1 and ELG2) on four bin layouts: bins of equal comoving width $\Delta d_{\rm com} = 62$ Mpc, and bins of fixed redshift width $\Delta z = 0.02$, 0.05 and 0.2. Bin i has mean redshift $\bar{z}_i$, and $D(z)$ is the linear growth factor, $D(0) = 1$. $w(a) = w_0 + w_a(1-a)$ is the dark-energy equation of state, with a the scale factor; Ω_m is the matter density today in units of the critical density, and A_s the amplitude of the primordial curvature power spectrum, sampled as $10^9 A_s$. A parameter's marginalized one-sigma error is σ. ${\rm FoM}(w_0, w_a) = [\det {\rm Cov}(w_0, w_a)]^{-1/2}$ is the inverse area of the marginalized (w_0, w_a) error ellipse, and ${\rm FoM}(\Omega_m, 10^9 A_s)$ is defined the same way. The reference is LSST Y1 3x2pt alone [7], on the same bands, with its own linear-bias lens sample.

The scale cut. Bin i is cut at $k_{\rm max}/D(\bar{z}_i)$: matter is less nonlinear at higher redshift, so the cut moves to higher wavenumber k as $1/D$, and every $k_{\rm max}$ in this note is the $z = 0$ value. The cut is the non-Limber UV-fraction cut of Saraivanov et al. [1], which follows DeRose and Chen [2]. Each band power is split into $C_\ell^{\rm IR}$, from the power spectrum below the bin's cut wavenumber, and $C_\ell^{\rm UV}$, from the power above it, and a band is kept while

$$\frac{|C_\ell^{\rm UV}|}{|C_\ell^{\rm UV}| + |C_\ell^{\rm IR}|} \; < \; 0.05. \tag{1}$$

The kept bands are intersected with the shipped mask, which stops every bin at $k = 0.1185\,{\rm Mpc}^{-1}$. The cut was built on the linear-bias model. The scan runs over $k_{\rm max} = 0.05, 0.075, \ldots, 0.2\,h/{\rm Mpc}$.

The covariance. The band-power covariance is Gaussian, computed for 14,000 square degrees with its inverse rescaled to the survey's 12,300, at the one-loop fiducial (Section 4): $b_{1,i} = B_g/D(\bar{z}_i)$, b_2 and b_{s^2} on their calibrations, every other term zero. Models A and B, with or without tilts, share this fiducial, so one covariance serves all of them. Its one-loop terms come from CosmoLike's C-FAST-PT tables, through the data vector's pair formula, in the Limber approximation; its low-ℓ non-Limber recompute is linear (Appendix S). Its non-Limber grid converges the adjacent thin-bin cross spectra to 2% and the diagonal to 0.4%. It takes massless neutrinos, so its signal sits 3% above the data vector ($m_\nu = 0.06$ eV) at every redshift. Against a linear-bias covariance the FoMs move by -0.4% to $+1.2\%$.

The linear-bias run. The notebook `EXAMPLE_EVALUATE_FISHER.ipynb`, the linear-bias run, fits the same priors with linear bias on the same kept band powers, so every difference from it is the bias model.

3 The linear-bias prior

The default model ties all bins of a group to one amplitude, $b_{1,i} = B_g/D(\bar{z}_i)$, with $b_{1,i}$ the linear bias of bin i and B_g one free amplitude per group g: a constant clustering amplitude $b_1 D$, as measured for DESI-like luminous red galaxies [14]. The tying makes thin bins powerful, but known physics moves each bin away from the group trend at the percent level (Appendix B), and a drift the model cannot express displaces the fitted cosmology by several σ (Section 6). One free bias per bin is accurate but spends data on the biases, 76 of them on $\Delta d_{\rm com} = 62$ Mpc. The prior below sits between the two: bins may leave the group trend, but neighbors move together.

The linear bias of bin i in group g is

$$b_{1,i} \;=\; \frac{B_g}{D(\bar{z}_i)}\,\big(1+\delta_i\big), \qquad \delta_i \;=\; s_g\,(\bar{z}_i - \bar{z}_g) + \varepsilon_i, \tag{2}$$

with B_g a free amplitude and s_g a free tilt per group (flat priors), $\bar{z}_g$ the mean of the group's bin redshifts, δ_i the fractional deviation from the group trend, and ε_i the residual around the group's line. The residuals carry a Gaussian prior correlated within a group and zero across groups,

$$\big\langle \varepsilon_i\,\varepsilon_j \big\rangle \;=\; \sigma_g^2\,\exp\!\left(-\,\frac{|\bar{z}_i - \bar{z}_j|}{L}\right), \tag{3}$$

with σ_g the prior width of one bin's residual and L the correlation length in redshift. σ_g sets how far one bin may wander from its group's line, L how far apart two bins must be before their residuals are independent, and together they set the rms jump between neighbors at separation Δz, $\sigma_g[2(1-e^{-\Delta z/L})]^{1/2}$ (Appendix A.2).

The tilt is free because B_g/D is not the evolution of a conserved population, whose bias obeys $b-1 \propto 1/D$ [9] (Appendix B.1), and every derived drift is close to a straight line across its group (Appendix C): the line absorbs coherent evolution, and the residual the curvature of a drift and local features of the selection (Appendix C.4). In the Fisher matrix the prior is one projected precision block per group that charges only the residual (Appendix C.5), so a group of one or two bins is free. Every group of the $\Delta z = 0.2$ layout is of this kind, and there the baseline, pessimistic and free priors coincide.

The four priors.

- **Growth-tied**: $b_{1,i} = B_g/D(\bar{z}_i)$, with no tilt and no residual; the default of the shipped yamls.
- **The baseline prior**: $\sigma_g = 3\%$ for BGS, 1% elsewhere, and $L = 0.2$. The BGS width holds the 2.4% rms curvature that a straight line leaves of the 22% drift of its $r < 19.5$ flux limit (Appendix E); the 1% is a stated hedge for unmodeled selection curvature, at least six times the 0.16% rms a conserved population leaves after its line (Appendix C.4). $L = 0.2$, comparable to the group spans, keeps the residual smooth across a group.
- **The pessimistic prior**: $\sigma_g = 3\%$ in every group and $L = 0.05$: local selection features as large as the largest derived curvature, on the redshift scale over which the 4000 Å break crosses a filter edge. That crossing is known, but its amplitude in bias is not measured (Appendix B.3).

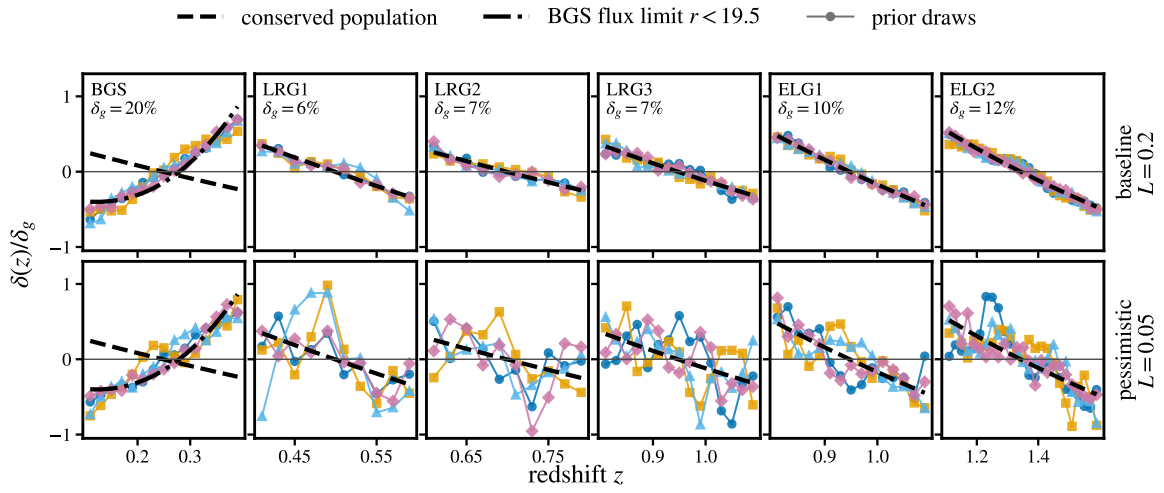


Figure 1: Draws of the baseline prior (top row) and the pessimistic prior (bottom row) on the $\Delta z = 0.02$ bins, one column per group, each column divided by the round constant δ_g of its top panel. Thick dashed: the conserved-population drift; thick dash-dot (BGS): the $r < 19.5$ flux-limit drift. Thin curves: four draws of $\delta(z)$ on the line of the group's reference drift, so draw and drift differ by the residual ε.

- **Free**: one $b_{1,i}$ per bin, each with a flat prior; the lower limit.

Figure 1 draws both priors around the known drifts. Under the baseline prior the neighbor rms of the residual at $\Delta z = 0.02$ is 0.44% (1.31% for BGS), and the known drifts sit well inside the smooth family; under the pessimistic prior it is 2.44% in every group: the histories keep their lines but gain local features nobody has measured.

4 Model A

Linear bias alone is not enough at $k_{\rm max} = 0.075\,h/\mathrm{Mpc}$: a linear-bias fit to a one-loop truth moves Ω_m by 1.7–1.9σ on $\Delta d_{\rm com} = 62$ Mpc under every linear-bias prior, and is safe only at $k_{\rm max} = 0.05\,h/\mathrm{Mpc}$ (`notes/linear_bias_kmax`). Model A (`notes/higher_order_bias`) keeps the per-bin linear bias of Eq. (2) and adds the one-loop terms; the real-space galaxy power spectrum of bin i is

$$P_{gg,i}(k) \;=\; b_{1,i}^2\,P_{\rm mm}(k) \;+\; P_{\rm loop}(k;\, b_{1,i}, b_{2,i}, b_{s^2,i}) \;+\; \alpha_0\,k^2 P_{\rm mm}(k) \;+\; \frac{\mathrm{SN}_0}{\bar{n}_i}, \qquad (4)$$

where $P_{\rm mm}$ is the nonlinear matter power spectrum and $P_{\rm loop}$ holds the one-loop terms. These carry the quadratic bias b_2, the response of the galaxy density to the square of the matter density contrast δ_m (the galaxy density contrast contains $b_2\,\delta_m^2/2$), and the tidal bias b_{s^2}, its response to the squared tidal field; the third-order bias b_3 is zero. α_0 is the k^2 counterterm, in $h^{-2}\,\mathrm{Mpc}^2$, which absorbs the sensitivity of the loops to small scales, and SN_0 the white-noise power beyond Poisson, in units of the bin's Poisson power $1/\bar{n}_i$, with $\bar{n}_i$ the bin's number density. The higher-order biases follow each bin's own linear bias through calibrations, plus

one offset per group:

$$b_{2,i} \;=\; b_2^{\rm L}(b_{1,i}) + \Delta_{2,g}, \qquad b_{s^2,i} \;=\; -\tfrac{4}{7}\,(b_{1,i}-1) + \Delta_{s,g}, \tag{5}$$

with $b_2^{\rm L}$ the separate-universe fit of Lazeyras et al. [3] and the second relation the co-evolution of a population formed with local bias [4].

Redshift space. Redshift-space distortions are linear (Kaiser), with each bin's redshift distribution convolved by the sampled velocity dispersion σ_v (the Fingers-of-God smearing). Two more terms depend on μ, the cosine of the angle to the line of sight. The counterterm α_2 adds, in the convention of the DESI DR1 full-shape analysis [6],

$$P_{\rm CT} \;=\; (b_1 + f\mu^2)\, f\, \alpha_2\, \mu^2 k^2 P_{\rm lin}, \tag{6}$$

with f the linear growth rate and $P_{\rm lin}$ the linear matter power spectrum; model A fixes $\alpha_2 = 0$. The stochastic power $\mathrm{SN}_2\, k^2\mu^2$ is counted in units of $f_{\rm sat}\sigma_{\rm sat}^2/\bar{n}_i$, with $f_{\rm sat}$ the satellite fraction and $\sigma_{\rm sat}$ the satellites' velocity dispersion in distance units: $(0.1, 7.8\,h^{-1}\mathrm{Mpc})$ for LRG, $(0.1, 2.9\,h^{-1}\mathrm{Mpc})$ for ELG, and the LRG values for BGS. Both stochastic terms project exactly, without the Limber approximation:

$$C_\ell^{ij}\big|_{\mathrm{SN}_0} = \mathrm{SN}_0\, P_{ij} \int d\chi\, \frac{W_i\, W_j}{\chi^2}, \qquad C_\ell^{ij}\big|_{\mathrm{SN}_2} = \mathrm{SN}_2\, f_{\rm sat}\sigma_{\rm sat}^2\, P_{ij} \int d\chi\, \frac{W_i'\, W_j'}{\chi^2}, \tag{7}$$

with χ the comoving distance, $W_i(\chi)$ the Fingers-of-God-convolved kernel of bin i, $W_i' = dW_i/d\chi$, and P_{ij} the Poisson power of the pair (Appendix M.1). Because $k^2\mu^2$ is the squared line-of-sight wavenumber, SN_2 is white noise differentiated along the line of sight, and its projection is that of SN_0 with each kernel replaced by its radial derivative.

Why model A carries SN_2. A thin bin's kernel derivative peaks at its two edges, with opposite signs. Adjacent bins share an edge, where one kernel rises as the other falls, so the projected SN_2 is anticorrelated between neighbors: normalized by the two autos, its cross term has a median of -0.50 over the 73 adjacent pairs of one tracer sample on $\Delta d_{\rm com} = 62$ Mpc, against $+0.05$ for SN_0. No white noise, per group or per bin, can mimic it. Without SN_2, model A broke at $k_{\rm max} = 0.05\,h/\mathrm{Mpc}$ on $\Delta d_{\rm com} = 62$ Mpc and before the first fitted cut on $\Delta z = 0.02$: an SN_2 truth at one prior width moved Ω_m by 1.3–1.7σ at 0.075, mostly through the adjacent cross spectra. The LRG groups carried that shift; SN_2 in BGS alone moved nothing, so the BGS unit does not matter, and in the ELG groups alone at most 0.23σ.

Priors and variants. The priors are Gaussian and centered on zero: widths 1 on Δ_2, 0.5 on Δ_s, 2 on SN_0, 5 on SN_2 (the DESI DR1 widths), and $0.5/k_{\rm max}^2 = 88.9\,h^{-2}\,\mathrm{Mpc}^2$ on α_0, the rule of the DESI 2024 full-shape analysis [5, 6] moved to this cut: $\alpha_0 k^2 = 0.5$ at $k = k_{\rm max}$. Since α_0 multiplies $P_{\rm mm}$, at bin i's own cut the width is a correction of $0.5/B_g^2$ to the linear galaxy power, the same in every bin of a group. **Model A** has one Δ_2, Δ_s, α_0, SN_0 and SN_2 per group, 30 parameters. In **model A + tilts**, Δ_2, Δ_s and α_0 also get a free straight line in redshift within each group, the kind of line s_g gives the linear bias in Eq. (2), with flat priors on the tilts: up to 48 parameters. SN_0 and SN_2 never tilt: their Poisson part is known bin by bin. Appendix M gives the implementation.

5 Model B and the reach of model A

Model B. Model B keeps model A's linear sector, calibrations and stochastic terms and adds the freedom a larger $k_{\rm max}$ calls for: α_2 (Eq. (6)) is free per group, and the offset widths double to 2 on Δ_2 and 1 on Δ_s. In model B + tilts α_2 gets the line as well: 36 parameters for model B, up to 60 for model B + tilts. In the scan the counterterm widths of both models follow $0.5/k_{\rm max}^2$ at every cut, DESI's $12.5\,h^{-2}\,{\rm Mpc}^2$ at $0.2\,h/{\rm Mpc}$. α_2 projects exactly, through the linear Kaiser transfer and its redshift-space row: it is linear to 10^{-12}, a brute-force integral over j_ℓ and j_ℓ'' agrees with it to 6×10^{-4}, and at $\alpha_2 = 12.5\,h^{-2}\,{\rm Mpc}^2$ it is 0.6–2.9% of the kept auto spectra. CosmoLike has no redshift-space one-loop terms; model B relies on the sampled σ_v and on α_2 for them.

Where the models break. A fit is given a truth that carries one term at one prior width, the same in every bin: $\alpha_2 = \pm12.5\,h^{-2}\,{\rm Mpc}^2$, at DESI's width, for model A, which lacks the term, and at model B's own width, $\pm0.5/k_{\rm max}^2$; ${\rm SN}_2 = \pm5$; the offsets (Δ_2, Δ_s) at one prior width of the fitted model, $\pm(1, 0.5)$ for model A and $\pm(2, 1)$ for model B; or the calibration drift of Section 6. The shift of the fit is the linear response of Appendix F.1, exact for α_2 and ${\rm SN}_2$, whose terms are linear, and first order for the offsets. A truth inside the fitted model's own prior—the offsets and ${\rm SN}_2$ for both models, and α_2 for model B—can move the fit by up to about 1σ without making its error bars dishonest, so its shifts are reported, never held to a threshold; the truths outside the prior are tested: the calibration drift, and, for model A, which fixes $\alpha_2 = 0$, the α_2 truth. A model holds at a cut when no tested truth moves (w_0, w_a), Ω_m or A_s by more than $\sigma/3$, there and at every smaller fitted cut; it breaks at the next cut. Figure 2c draws the largest tested shift in two scenarios: the baseline prior with models A and B, and the pessimistic prior with models A and B + tilts. Each variant breaks at the same cut under the baseline, pessimistic and free priors.

- Without tilts model A holds to $0.075\,h/{\rm Mpc}$ on $\Delta z = 0.02$ (α_2 moves (w_0, w_a) by 1.28σ at 0.1) but only to 0.05 on $\Delta d_{\rm com} = 62$ Mpc, where α_2 moves A_s by 0.334σ at 0.075, just above $\sigma/3$, and to 0.05 on $\Delta z = 0.05$ and 0.2, ended by the calibration drift (0.46 and 0.39σ at 0.075). Model B, whose α_2 lies inside its prior, is tested by the drift alone: it holds to 0.075 on the two thin layouts, where the drift moves (w_0, w_a) by 0.81 and 0.87σ at 0.1, and to 0.05 on $\Delta z = 0.05$ and 0.2 (0.43 and 0.35σ at 0.075). From 0.1 on the largest tested shift is 0.53–2.8σ, and from 0.125 on the drift sets it in every fit.

- The tilts absorb the calibration drift exactly, so α_2, which model A fixes to zero, is the only truth that tests model A + tilts. It holds to 0.075 on $\Delta z = 0.02$ (α_2 moves (w_0, w_a) by 1.32σ at 0.1); to 0.05 on $\Delta d_{\rm com} = 62$ Mpc (A_s by 0.35σ at 0.075); to 0.15 on $\Delta z = 0.05$ (A_s by 0.54σ at 0.175); and at every cut on $\Delta z = 0.2$, whose wide bins barely see α_2 (at most 0.05σ).

- Model B + tilts holds at every cut on every layout, by construction: its only truth outside its prior is the drift, which its tilts absorb exactly (shifts of order $10^{-12}\sigma$), so this pass carries no information on its own. Its accuracy rests on the reported in-prior shifts—up to 0.99σ at 0.075 on $\Delta d_{\rm com} = 62$ Mpc and 0.33–1.09σ on the three thinner layouts from 0.1 on, set mostly by α_2, and at most 0.22σ on $\Delta z = 0.2$—and on its honest error bars (Section 6).

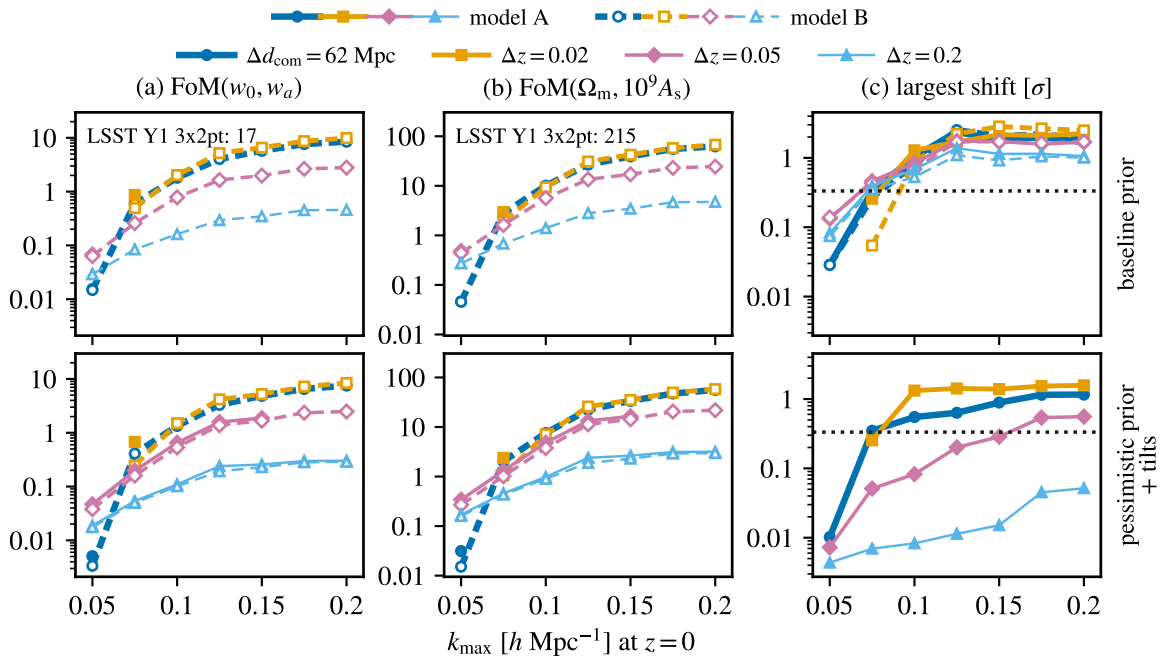


Figure 2: DESI C_ℓ^{gg} alone against $k_{\rm max}$ in two scenarios: the baseline prior with models A and B (top) and the pessimistic prior with models A and B + tilts (bottom). (a) FoM(w_0, w_a); (b) FoM($\Omega_m, 10^9 A_s$); (c) the largest shift of a fit over (w_0, w_a), Ω_m and A_s and over the truths outside its prior (the calibration drift, and α_2 for model A, which fixes it to zero); the shifts of the truths inside the prior are reported in the folder's `data.json`. Color, line width and marker: the layouts. Solid, filled: model A, in (a) and (b) up to its break; dashed, open: model B; model B + tilts is absent from (c), since its tilts absorb the drift exactly. Gray long-dashed in (a), text in (b): LSST Y1 3x2pt; dotted: $\sigma/3$. Omitted: the free prior. On $\Delta z = 0.02$ the cut at 0.05 keeps no band. $k_{\rm max}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

With SN_2 in the fit, an SN_2 truth at one prior width lies inside both models' priors, so it is reported, never tested; it moves no parameter by more than 0.39σ at any cut.

The offset truth stays inside model A's priors. If the calibrations instead scatter as widely as model B allows, $\pm(2, 1)$, model A's tighter offset priors shift its largest score by 0.19–0.44σ at 0.075 and by 0.45–0.98σ from 0.1 up: a diagnostic of the prior, outside the break point. Likewise, when α_2 scatters over model B's whole prior, seven times $12.5\,h^{-2}\,\mathrm{Mpc}^2$ at 0.075, model A's error bars on the two thin layouts at that cut are too small by factors of 1.5–3.3 and cover the truth in 13–50% of universes instead of 68%; they recover as the prior narrows with $k_{\rm max}$, and on the wide layouts they are too small by at most 11% at every cut.

The FoM against $k_{\rm max}$. Figure 2a and b draw the FoMs of the two scenarios, model A up to its break and model B at every cut, with LSST Y1 3x2pt (17.26 and 214.9) for reference. The baseline scenario gives the larger FoMs at every cut, FoM(w_0, w_a) = 1.09 and 0.866 for model A at 0.075 on the thin layouts and 8.47 and 9.92 for model B at $0.2\,h/\mathrm{Mpc}$, with FoM($\Omega_m, 10^9 A_s$) at most 67.6, but fails the break test from 0.075 on. On the pessimistic prior, model A + tilts

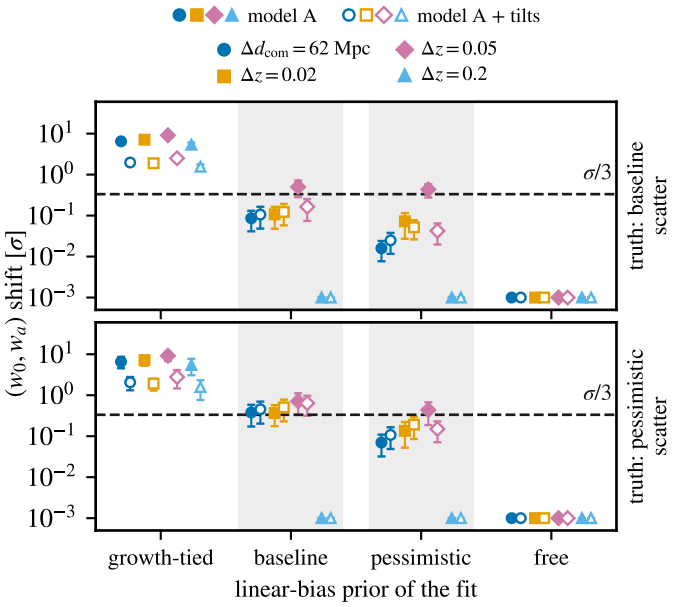


Figure 3: The (w_0, w_a) shift of each fit in units of its own ellipse, for truths with a residual drawn from the baseline prior (top) or the pessimistic prior (bottom), without data noise. Filled: model A; open: model A + tilts; colors and markers: the layouts. Points: the mean shift; bars: 16th–84th percentiles. Gray columns: the baseline and pessimistic priors. Dashed: $\sigma/3$. Points at 10^{-3}: exact zeros. $k_{\rm max} = 0.075\,h/{\rm Mpc}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

gives FoM$(w_0, w_a) = 1.01$ and 0.675 on the thin layouts at 0.075. Model B + tilts, the adopted model, gives FoM$(w_0, w_a) = 0.411$, 0.244, 0.160 and 0.0496 at 0.075 and 7.54, 8.43, 2.48 and 0.288 at 0.2 (layout order), and FoM$(\Omega_m, 10^9 A_s) = 55.9$, 58.2, 21.8 and 2.97 at 0.2. The free prior, and the model A FoMs beyond the break, which extending model A would claim, are in `plots/fom_kmax_models/data.json`.

6 The accuracy test

No mock catalogs and only two derived drift laws are available, so this note does not defend one prior. It tests each prior against random truths at $k_{\rm max} = 0.075\,h/{\rm Mpc}$ under model A and quotes the FoM only for the priors that pass.

Random truths. Each truth sets every bin's linear bias to its group's known drift (the $r < 19.5$ flux limit in BGS, a conserved galaxy population elsewhere; Appendix F.2) plus a residual drawn from the baseline or the pessimistic prior, 2000 truths per family, with b_2 and b_{s^2} following b_1 through Eq. (5). Every prior fits every truth under model A, with and without tilts, centered on the growth-tied shape, to first order in the truth (Appendix F). Figure 3 draws the (w_0, w_a) shifts. Where a fit's prior equals the truth's scatter, the mean shift is not a bias: that fit's ellipse already includes this scatter (Appendix F.4).

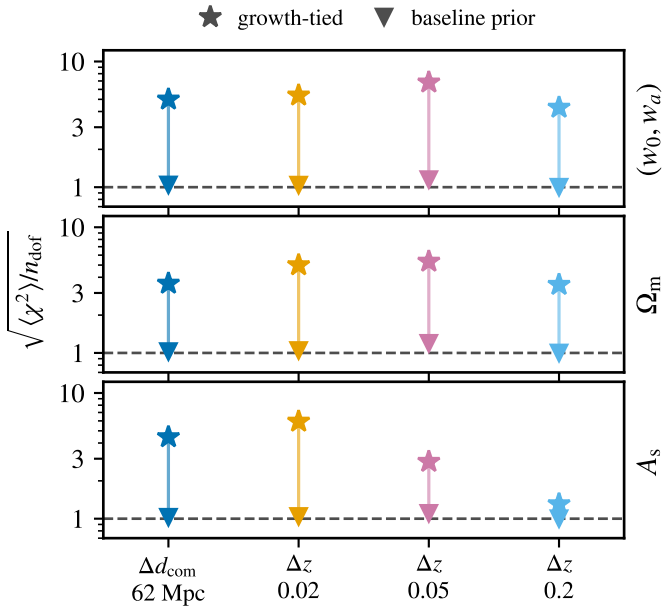


Figure 4: How well each fit's quoted error bar covers where the fit actually lands: $\eta = \sqrt{\langle\chi^2\rangle/n_{\rm dof}}$ of (w_0, w_a), Ω_m and A_s under the pessimistic scatter, model A. Stars: the growth-tied prior; down-triangles: the baseline prior; dashed: $\eta = 1$. A one-parameter 1σ bar covers the truth in a fraction $2\Phi(1/\eta) - 1$ of universes, Φ the Gaussian cumulative distribution. Omitted: the free prior (exactly 1) and the pessimistic prior (at most 1.08). $k_{\rm max} = 0.075\,h/{\rm Mpc}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

The statistic. Each fit is scored by

$$\sqrt{\langle\chi^2\rangle/n_{\rm dof}}, \qquad \chi^2 = \Delta p^{\mathsf{T}}\, \mathsf{C}_p^{-1}\, \Delta p, \tag{8}$$

with Δp the recovered minus the true parameters, C_p the fit's marginalized covariance, $n_{\rm dof} = 2$ for $p = (w_0, w_a)$ and 1 for Ω_m or A_s, and $\langle\cdot\rangle$ the exact average over the data noise and the bias history. It measures how well each fit's quoted error bar covers where the fit actually lands. Its value, written η, is 1 for error bars that cover the truth as often as they claim; above 1 they are too small by that factor, and below 1 conservative. It scores the parameters, not the χ^2 of the data, and for one parameter it converts into the fraction of universes in which the 1σ bar covers the truth (Appendix F.4).

The result of the test. Figures 3 and 4 order the priors.

- *The growth-tied prior fails under every truth and is discarded.* The conserved-population drift alone moves its (w_0, w_a) by several σ (Appendix N), and under the pessimistic scatter its Ω_m bar covers the truth in only 0.4–12% of universes. Under model A + tilts, whose flat-prior tilts follow part of a drift, η of (w_0, w_a) is still 1.5–2.4.

- *The baseline prior is honest only if the residual is as smooth as it assumes.* Under its own scatter η is 1.00–1.11. Under the pessimistic scatter it stays below 1.05 on every

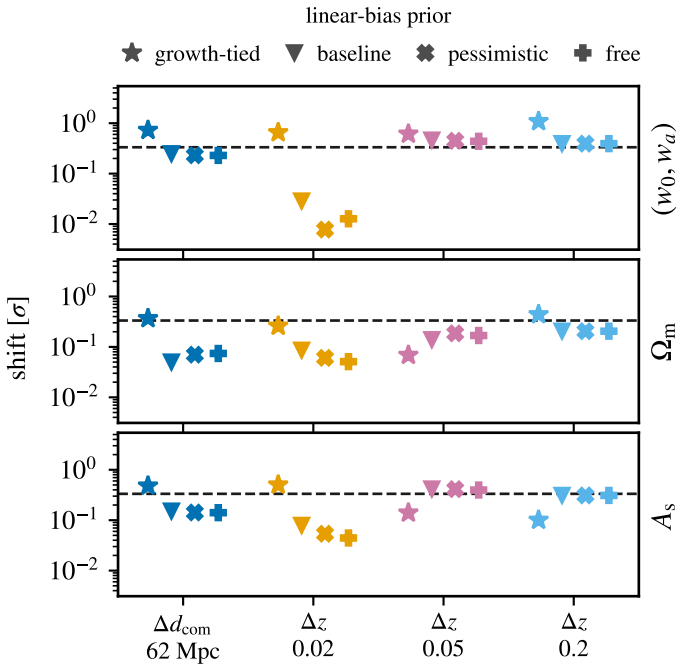


Figure 5: The shift of a model A fit under the calibration drift (Δ_2 and Δ_s changing linearly by 1 and 0.5 across each group): (w_0, w_a) in units of the fit's ellipse, Ω_m and A_s in their own σ. Markers: the four linear-bias priors; dashed: $\sigma/3$. Omitted: model A + tilts, which absorbs the drift exactly. $k_{\max} = 0.075\,h/\mathrm{Mpc}$ quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

layout but $\Delta z = 0.05$, where η reaches 1.19: the Ω_m bar covers the truth in 67, 66 and 60% of universes on $\Delta d_{\mathrm{com}} = 62$ Mpc, $\Delta z = 0.02$ and 0.05, and the A_s bar in 67, 66 and 64%. Model A + tilts does not change this.

- *The pessimistic prior is honest under both scatters*: $\eta \leq 1.08$ in every score, and 0.99–1.04 in (w_0, w_a) under baseline scatter. The 1.08 is the BGS flux-limit drift on $\Delta z = 0.05$, which this prior leaves just above $\sigma/3$ (Appendix N). The free prior gives exactly 1.

A drift of the calibrations. The random truths move only the linear bias. A third truth tests the calibrations: it drifts the offsets linearly across each group, Δ_2 from about -0.5 to $+0.5$ and Δ_s from about -0.25 to $+0.25$, with b_1 at its fiducial (Appendix M.4). Under model A (Figure 5) it moves (w_0, w_a) by more than $\sigma/3$ under every prior on the wide layouts, $\Delta z = 0.05$ and 0.2, and A_s as well on $\Delta z = 0.05$; on the thin layouts only the growth-tied prior exceeds $\sigma/3$. Model A + tilts absorbs the drift exactly, since its tilts have the drift's shape; a nonlinear drift of the calibrations within a group is untested. The drift is also one of the tested truths of Section 5.

The test at every cut. The test above fixes $k_{\max} = 0.075\,h/\mathrm{Mpc}$ and model A. Figure 6 repeats it at every cut under model B + tilts, the adopted model, for the growth-tied prior against the pessimistic prior, with the same truths: the known drifts plus a residual drawn

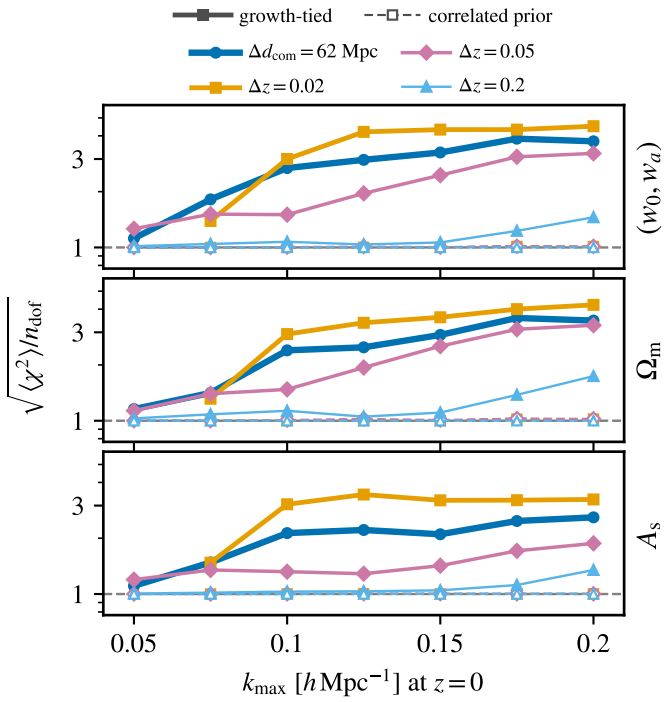
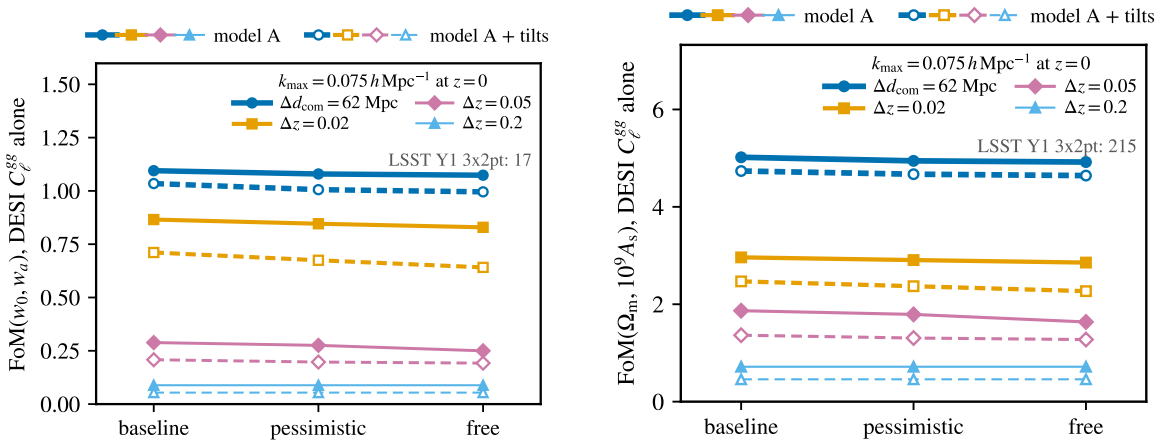


Figure 6: The accuracy test at every cut: $\eta = \sqrt{\langle\chi^2\rangle/n_{\rm dof}}$ of (w_0, w_a), Ω_m and A_s (one panel each, one shared logarithmic scale) for model B + tilts fitted with the growth-tied prior (solid, filled markers) and with the correlated prior, here the pessimistic prior (thin dashed, open markers), to truths that carry the known drifts plus a residual drawn from the pessimistic prior. Color, line width and marker: the layouts. Gray dashed: $\eta = 1$, honest error bars. The pessimistic prior stays at $\eta \leq 1.03$ at every cut; the growth-tied prior reaches 3.3–4.5 on the three thinner layouts at $0.2\,h$/Mpc, where its Ω_m bar covers the truth in 2–9% of universes. $k_{\rm max}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

from the pessimistic prior. The pessimistic prior keeps $\eta \leq 1.03$ at every cut and on every layout. Under the growth-tied prior η grows from 1.0–1.5 at the smallest fitted cut to a largest score of 3.3–4.5 on the three thinner layouts at $0.2\,h$/Mpc, where its Ω_m bar covers the truth in 2–9% of universes, and 1.7 on $\Delta z = 0.2$.

7 The figure of merit at $k_{\rm max} = 0.075\,h$/Mpc

A FoM means something only for a prior that passes the accuracy test: the pessimistic and free priors, and the baseline prior only if the bias residual is as smooth as it assumes. Figure 7 draws these three priors under model A, with and without tilts. Both FoMs fall from the baseline prior to free, except on $\Delta z = 0.2$; the thin layouts carry most of the information, and the tilts cost little on them and more on the wide layouts. DESI C_ℓ^{gg} alone sits well below LSST Y1 3x2pt in both FoMs. Appendix Q sets these FoMs next to linear bias and exact calibrations, neither of which is a result.



(a) (b)

Figure 7: **(a)** FoM(w_0, w_a) and **(b)** FoM$(\Omega_m, 10^9 A_s)$ of DESI C_ℓ^{gg} alone under the baseline, pessimistic and free priors. Color, line width and marker: the layouts; solid, filled: model A; dashed, open: model A + tilts. The text in each panel quotes the LSST Y1 3x2pt value, far above the DESI points. $k_{\max} = 0.075\,h/\mathrm{Mpc}$ quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

Result. At this cut, model A + tilts on the pessimistic prior stays honest under every truth tested, and the free prior is the lower limit. Model A on the baseline prior shows what better knowledge of bias evolution would buy, if each bin's residual is no larger and no rougher than the baseline prior and the calibrations hold within a group up to one constant offset. Figure 2 carries both to other cuts, where model B + tilts on the pessimistic prior is the adopted model (Section 5).

A The models

This appendix first builds the prior without the tilt of Eq. (2): the kernel model, in which the exponential kernel describes the whole deviation δ_i of a bin from its group's trend. Appendix C adds the free tilt, and the kernel then describes the residual ε_i around each group's line. In the appendices b_i is the linear bias $b_{1,i}$ of bin i.

A.1 Growth-tied, free, and correlated biases

The growth-tied model sets $b_i = B_g/D(\bar{z}_i)$, with one free amplitude B_g per group. The kernel model multiplies it by a fractional deviation δ_i,

$$b_i \;=\; \frac{B_g}{D(\bar{z}_i)}\,\big(1+\delta_i\big), \tag{9}$$

$$\big\langle \delta_i\,\delta_j \big\rangle \;=\; \sigma_g^2\,\exp\!\left(-\,\frac{|\bar{z}_i-\bar{z}_j|}{L}\right), \tag{10}$$

with B_g still free (flat prior) and the prior zero across groups. In the Fisher matrix the convenient parameters are the per-bin amplitudes

$$\beta_i \;\equiv\; B_g\,\big(1+\delta_i\big), \qquad \text{fiducial } \beta_i = B_g, \tag{11}$$

so that $b_i = \beta_i/D(\bar{z}_i)$. Since $\beta_i - B_g = B_g\,\delta_i$, the prior covariance of the β_i within one group is

$$\mathsf{C} \;=\; \sigma_\beta^2\,\mathsf{R}, \qquad \sigma_\beta \;=\; \sigma_g\,B_g, \qquad \mathsf{R}_{ij} \;=\; \exp\!\left(-\,\frac{|\bar{z}_i-\bar{z}_j|}{L}\right), \tag{12}$$

with σ_β the spread of one amplitude, in units of a bias, and R the dimensionless correlation matrix.

A.2 What σ_g and L mean

The diagonal of the kernel gives the wander of a single bin, $\mathrm{Var}(\delta_i) = \sigma_g^2$. The correlation coefficient of two bins at separation $\Delta z = |\bar{z}_i - \bar{z}_j|$ is

$$r \;=\; \frac{\langle \delta_i \delta_j \rangle}{\sqrt{\langle \delta_i^2 \rangle\,\langle \delta_j^2 \rangle}} \;=\; e^{-\Delta z/L}, \tag{13}$$

set by the separation and L alone. The jump between the two bins has variance

$$\mathrm{Var}(\delta_i - \delta_j) \;=\; 2\,\sigma_g^2\left(1 - e^{-\Delta z/L}\right) \;\approx\; \frac{2\,\sigma_g^2\,\Delta z}{L} \quad (\Delta z \ll L). \tag{14}$$

At small separations the jump variance grows linearly with Δz, as for a random walk; only beyond L does the finite wander pull the deviation back. For neighbors at $\Delta z = 0.02$ with $L = 0.05$, $r = e^{-0.4} = 0.670$ and the exact rms jump is $0.81\,\sigma_g$; the small-separation form gives $0.89\,\sigma_g$, because $\Delta z/L = 0.4$ is not small.

A.3 The free group amplitude and the projection

A plain Gaussian prior with covariance C would charge for moving all β_i of a group together, but that uniform shift is the free amplitude B_g. The repair removes the uniform direction: with $\mathbf{1}$ the ones vector of the group's n bins, the prior precision of a group's amplitudes is

$$\mathsf{P} \;=\; \mathsf{C}^{-1} \;-\; \frac{\mathsf{C}^{-1}\mathbf{1}\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}}{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}}. \tag{15}$$

$\mathsf{P}\,\mathbf{1} = 0$, so a uniform shift costs nothing; a pure shape change keeps the penalty of the plain prior; P is positive semidefinite with rank $n-1$; and P is exactly the prior obtained by sampling B_g with a flat prior and marginalizing it. Appendices G, H and I show these for two bins, by marginalization, and for n bins. Across groups the prior is zero, so the full prior precision is block diagonal.

B Which L and σ_g are physical

One case of bias evolution can be solved in closed form: a population whose members are conserved. This appendix derives it, measures its deviation from the growth-tied trend, lists what breaks conservation in a selected sample, and weighs the published measurements.

B.1 A conserved population: the bias continuity equation

Take a sample whose membership never changes, with no velocity bias, in linear theory. The matter overdensity δ_m and the galaxy-count overdensity $\delta_{\rm gal}$ then obey the same continuity equation with the same velocity field, so the excess clumping $\delta_{\rm gal} - \delta_m$ is frozen (Appendix D). With b_c the linear bias at a reference redshift z_c and $\delta_m \propto D$, the frozen difference gives [9]

$$b(z) - 1 \;=\; \big(b_c - 1\big)\,\frac{D(z_c)}{D(z)}\,: \tag{16}$$

the excess set at formation stays fixed while the matter clustering grows, so the bias decays toward one. The growth-tied shape freezes a different quantity: constant $b\,D$ means a galaxy clustering amplitude $b\,\delta_m$ that does not grow, which describes a selection that picks, at each redshift, a different and rarer set of objects.

B.2 The deviation from the growth-tied trend

Place the reference at a group's midpoint z_c, with $b_c = B/D(z_c)$. The ratio of Eq. (16) to the growth-tied bias $b_c\,D(z_c)/D(z)$ gives the deviation of Eq. (9),

$$\delta(z) \;=\; \frac{D(z)/D(z_c) - 1}{b_c}, \tag{17}$$

zero at the midpoint, positive at lower redshift and negative at higher redshift, and nearly linear across a group because D is smooth. In flat ΛCDM with $\Omega_m = 0.3$ and the project's fiducial amplitudes it is $+5.3\%$ and -4.9% at the ends of BGS ($B = 1.34$, $0.1 < z < 0.4$, $b_c = 1.5$), $+2.4\%$ and -2.2% at the ends of LRG1 ($B = 1.70$, $0.4 < z < 0.6$, $b_c = 2.2$), and $+2.6\%$ and -2.3% at the ends of LRG3 ($B = 1.70$, $0.8 < z < 1.1$, $b_c = 2.7$). Between neighbors

it changes by 0.3–0.7% at $\Delta z = 0.02$ and 0.8–1.7% at $\Delta z = 0.05$: a smooth tilt, coherent over the whole group. For comparison, the neighbor rms of Eq. (14) at $\Delta z = 0.02$ (and 0.05) is 2.4% (3.4%) for $\sigma_g = 3\%$ and $L = 0.05$, and 0.44% (0.67%) for $\sigma_g = 1\%$ and $L = 0.2$. Figure 8 draws these shapes: the conserved curve is flatter than B_g/D in every group, while the BGS flux-limit curve is much steeper, and convex.

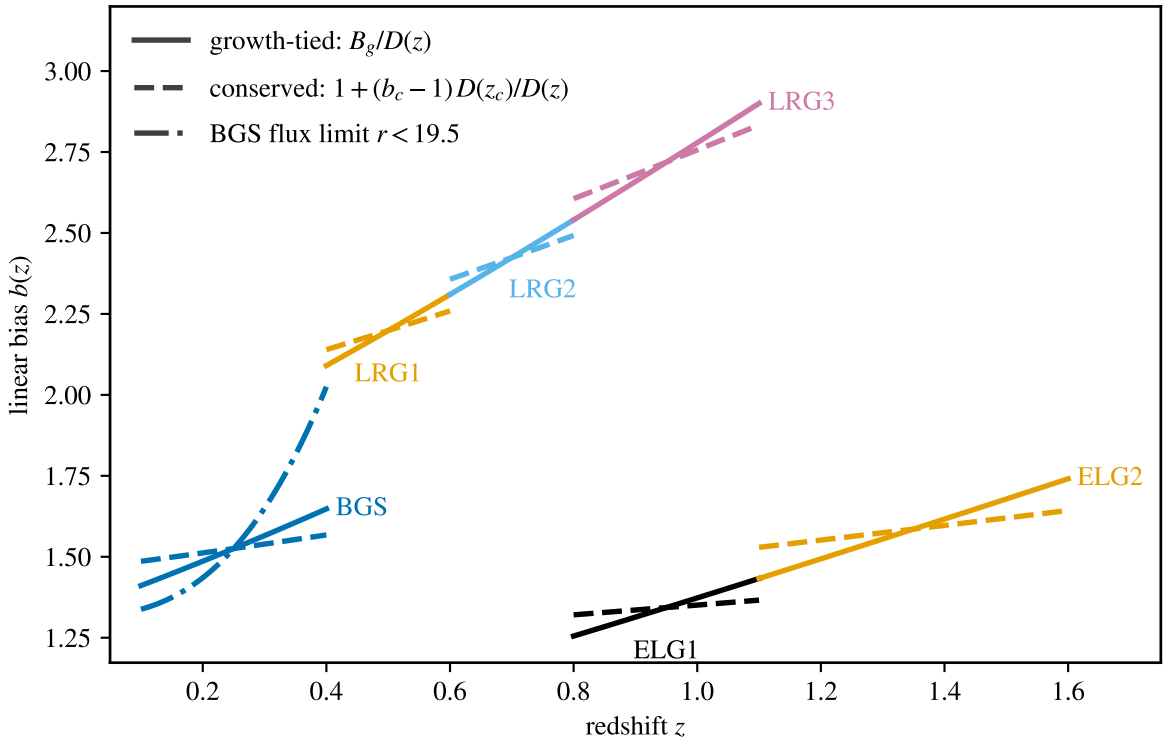


Figure 8: The bias $b(z)$ of every group. Solid: the growth-tied shape $B_g/D(z)$. Dashed: a conserved population anchored at the group midpoint (Eq. (16)). Dash-dot, on BGS only: the $r < 19.5$ flux-limit bias, anchored at the same midpoint.

B.3 What breaks conservation in a selected sample

- **Threshold drift.** A selection holds a roughly constant comoving density until the flux limit bites; beyond, it keeps only its brighter, more biased members. The change is smooth, with L comparable to the group width.
- **Color-selection features.** A spectral feature crossing a filter edge moves a color cut quickly in redshift. The sharpest is the 4000 Å break: a filter edge 100–200 Å wide maps to $\Delta z \approx 0.03$–0.05. The crossing is known; its amplitude in bias is unmeasured.
- **Mergers** turn two members into one.
- **Emission-line selection.** An ELG belongs to the sample while its lines are strong, so membership follows the star-formation phase.

B.4 What the measurements say

Each statement is tagged *measured* (read from the cited source), *derived* (computed here from measured inputs) or *assumed*. Equation (16) is from Fry [9]; Tegmark and Peebles [10] rederive it.

Massive red galaxies follow the conserved shape. SDSS and 2SLAQ LRGs at $0.19 < z < 0.55$, matched in color and luminosity, show no significant evolution of the clustering amplitude and merge at $7.5 \pm 2.3\%$ over that range, about 2.4% per Gyr [11] (measured). The BOSS progenitors of the SDSS LRGs merge at $2 \pm 1.5\%$ per Gyr and cluster as the continuity equation predicts [12], and CMASS samples of fixed comoving density at $0.43 < z < 0.7$ cluster like a conserved population [13] (measured). The bias of DESI-like LRGs in photometric bins of $\Delta z \approx 0.1$ stays within $\pm 2.3\%$ of $b = 1.5/D$ below $z = 0.83$, with the highest bin $6.4 \pm 6.4\%$ high [14] (measured biases, residuals derived; Figure 9).

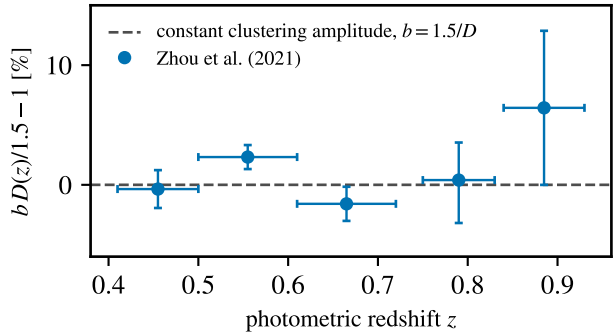


Figure 9: The measured bias of DESI-like LRGs [14] as a fractional residual around $b = 1.5/D$, with D at each photometric bin's center (flat ΛCDM, $\Omega_m = 0.3$); horizontal bars: the bin edges.

The selections drift. The DESI LRG cuts hold the comoving density near $5 \times 10^{-4}\,(h/\mathrm{Mpc})^3$ at $0.4 < z < 0.8$ and hand over to a fixed flux limit beyond [15] (measured). That limit raises the luminosity threshold by 0.27 magnitudes per $\Delta z = 0.1$ at $z \approx 1$; times the threshold-bias slope of SDSS luminosity-threshold samples, $b\,(\sigma_8/0.8) = 1.06 + 0.21\,(L_{\mathrm{th}}/L_*)^{1.12}$ [20] (0.5–1.0 in bias per magnitude near LRG luminosities; measured), it drifts the LRG3 bias by 6–11% per $\Delta z = 0.1$ on top of B/D (derived, from an r-band relation outside its calibrated range, so only the direction and order of magnitude hold). BGS Bright is magnitude limited at $r < 19.5$ [23] (measured), which drives a 10–25% bias drift across the group (derived; Appendix E).

The ELG bias shape is stable. HOD fits of the DESI One-Percent ELGs in $0.8 < z < 1.1$ and $1.1 < z < 1.6$ agree within one sigma [25] (measured); at these host masses the halo bias depends weakly on mass, so the implied drift is at the percent level (derived, slope assumed). The [OII] doublet loses signal-to-noise in windows $\Delta z \approx 0.001$–0.01 wide where it lands on sky lines, especially at $z > 1.5$ [24] (measured): the one short-separation mechanism with a measured location, with an assumed bias amplitude of 1–2%.

Nothing resolves bias below $\Delta z \approx 0.1$. The CMB-lensing tomography of the same samples [17, 18, 19], on the photometric samples of Zhou et al. [16], uses bins of about 0.15 with one free bias per bin and finds smooth models that fit. Below $\Delta z \approx 0.1$ this note rests on the continuity equation and the selection arguments.

C The memory model

C.1 Why a second model

Every deviation above except the sky-line windows is a monotonic drift: the conserved-population tilt (about 1% per $\Delta z = 0.05$), the LRG3 magnitude-limit drift (6–11% per $\Delta z = 0.1$) and the BGS flux-limit drift (10–25% across the group). The kernel model admits such a drift only at a price: to pass a 20% drift with $L \leq 0.1$ it takes $\sigma_g = 10$–15%, which also allows about 10% scatter between neighbors that no identified mechanism produces. One σ_g controls both, so the kernel cannot say "large smooth drift, small local scatter". The memory model separates the two.

C.2 Why the kernel forgets

At $L = 0.05$, bins 0.15 apart share $e^{-3} \approx 0.05$ of their deviation. With $r_k = e^{-(\bar{z}_{k+1} - \bar{z}_k)/L}$ the correlation of neighbors k and $k+1$, the best guess of the next bin given all earlier ones is (Appendix I.2)

$$\mathrm{E}\big[\delta_{k+1} \mid \delta_1, \ldots, \delta_k\big] \;=\; r_k\, \delta_k : \tag{18}$$

only the nearest neighbor matters, and the influence of any one bin fades along the chain.

C.3 The model, and why it remembers

The memory model adds a tilt per group and splits the deviation into a line and a residual,

$$\delta_i \;=\; s_g\, v_i + \varepsilon_i, \qquad v_i \;=\; \bar{z}_i - \bar{z}_g, \tag{19}$$

with s_g a free tilt (flat prior, like B_g) and ε_i carrying the kernel of Eq. (10): the model of Eqs. (2) and (3). Giving the slope a Gaussian prior of variance τ^2 first (the flat prior is $\tau \to \infty$), the deviations have covariance

$$\mathrm{Cov}(\delta_i, \delta_j) \;=\; \tau^2\, v_i\, v_j \;+\; \sigma_g^2\, e^{-|\bar{z}_i - \bar{z}_j|/L}. \tag{20}$$

The new term does not decay with separation. Far beyond L the correlation coefficient tends to

$$\rho_{ij} \;\longrightarrow\; \frac{\tau^2 v_i v_j}{\sqrt{\big(\tau^2 v_i^2 + \sigma_g^2\big)\big(\tau^2 v_j^2 + \sigma_g^2\big)}} \;\longrightarrow\; \pm 1 \quad \text{when } \tau|v| \gg \sigma_g : \tag{21}$$

the two ends of a group remember each other through the shared slope. The best guess of the next bin becomes

$$\mathrm{E}\big[\delta_{k+1} \mid \delta_1, \ldots, \delta_k\big] \;=\; \hat{c} + \hat{s}_g\, v_{k+1} \;+\; r_k\, \big(\delta_k - \hat{c} - \hat{s}_g\, v_k\big), \tag{22}$$

with $(\hat{c}, \hat{s}_g)$ the level and slope of the best straight line through all past bins, weighted by C^{-1}: every past bin enters through the fitted line, and only the wiggle around it stays Markov (the

universal-kriging predictor; Appendix J). Figure 10 shows the difference on the BGS bins at $\Delta z = 0.02$: kernel draws wide enough to admit the BGS drift zigzag and forget, while memory draws are lines with small wiggles, among which the drift looks typical.

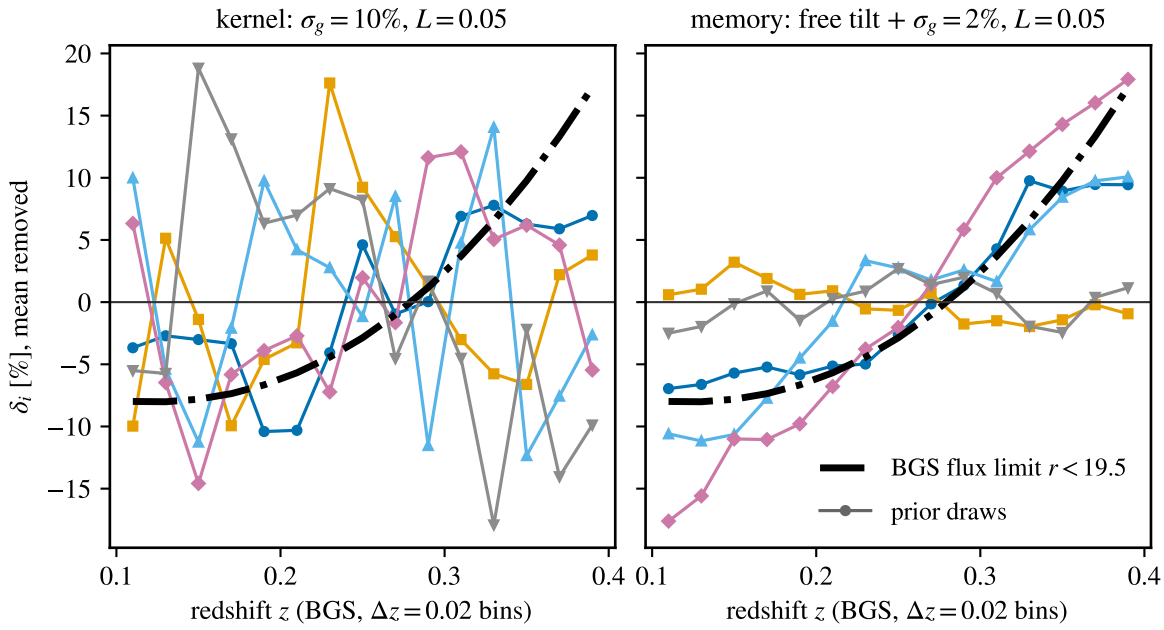


Figure 10: Random prior draws of the deviations of the 15 BGS bins at $\Delta z = 0.02$, each draw's mean removed (B_g is free). Thick dash-dot: the BGS flux-limit drift. Left: the kernel model. Right: the memory model, whose tilt is free in the model and drawn here, for display only, from a Gaussian of width 0.6 per unit redshift.

C.4 Why the memory model differs physically

A selected sample changes along a group in two ways. Coherent changes move every bin the same way: a threshold that rises with distance, the debiasing of a conserved population. Local changes move bins independently and are forgotten over a short range: the sky-line windows and filter-edge crossings of Appendix B.3. The memory model gives the first kind to the tilt and the second to the kernel. One line suffices for the coherent kind, because the drifts are smooth functions of $D(z)$ and the distance modulus. The best equal-weight line through each group's conserved deviation at the $\Delta z = 0.05$ bin centers (derived; amplitudes 1.34 for BGS, 1.70 for the LRGs and 0.84 for the ELGs) removes deviations that span 3–11% and leaves 0.02–0.16% rms, at most 0.27% (ELG2). The BGS flux-limit drift is larger and convex: it spans 22.3%, and its best line leaves 2.4% rms and 3.3% at most (Appendix E). Figure 11 shows both. In the memory model σ_g is the departure from the group's own line, curvature plus local scatter: about 0.1% for a conserved population, 1–2% for local selection features, 2–3% for the BGS curvature.

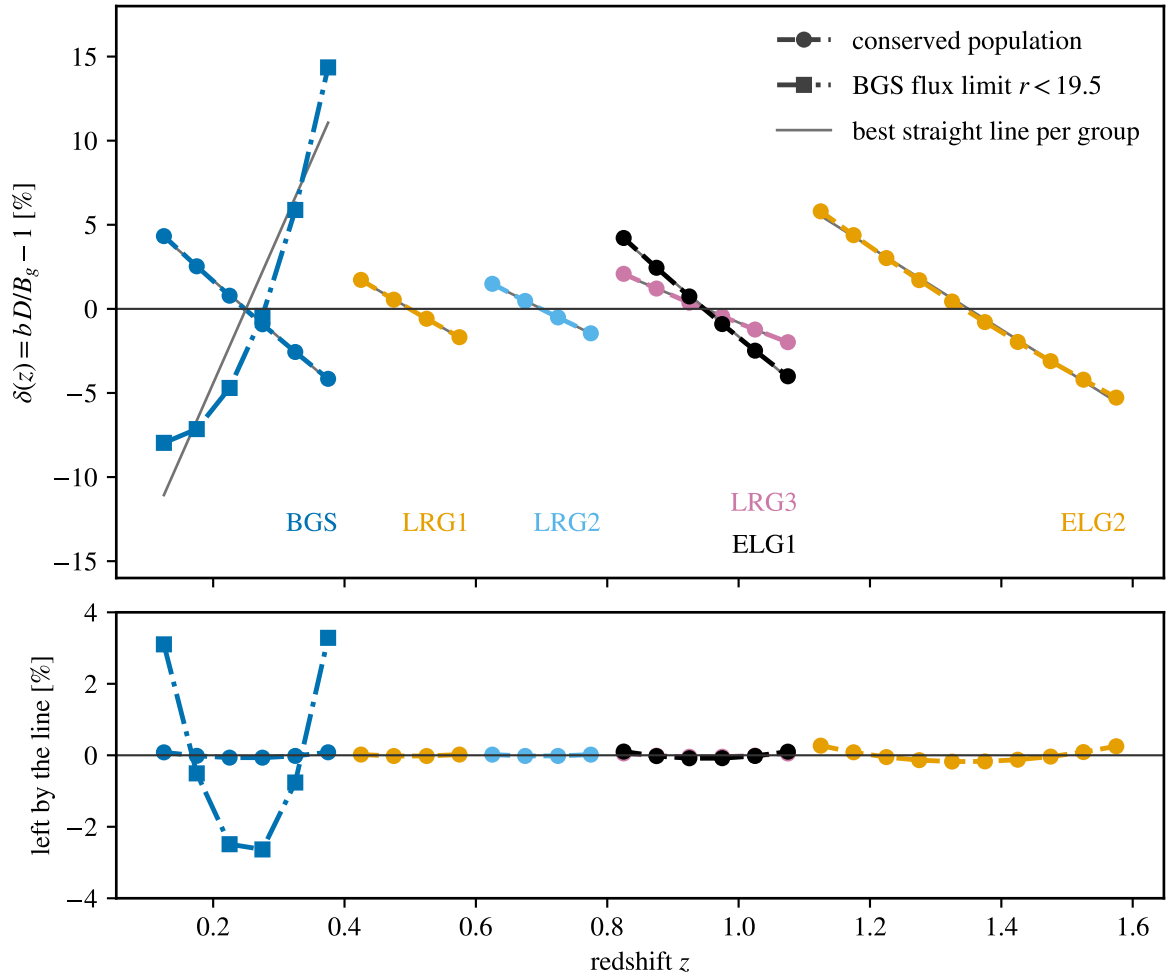


Figure 11: Top: the deviation $\delta(z) = b\,D/B_g - 1$ at the $\Delta z = 0.05$ bin centers, with B_g set so the mean deviation vanishes. Dashed with circles: the conserved population, every group. Dash-dot with squares: the BGS flux limit. Thin gray: the best straight line per curve. Bottom: what the line leaves.

C.5 What the prior charges

With flat priors on the line's level and slope, the prior precision of a group's bin amplitudes is (Appendix J)

$$\mathsf{P}_{\text{mem}} \;=\; \mathsf{Q} - \mathsf{Q}\,\mathsf{U}\big(\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\big)^{-1}\mathsf{U}^{\mathsf{T}}\mathsf{Q}, \qquad \mathsf{Q} = \mathsf{C}^{-1}, \qquad \mathsf{U} = [\,\mathbf{1},\; v\,], \tag{23}$$

the construction of Eq. (15) with one more column. Its penalty is

$$\delta^{\mathsf{T}}\mathsf{P}_{\text{mem}}\,\delta \;=\; \min_{c,\,s_g} \big(\delta - c\,\mathbf{1} - s_g\,v\big)^{\mathsf{T}}\mathsf{Q}\big(\delta - c\,\mathbf{1} - s_g\,v\big)\,: \tag{24}$$

the prior subtracts the best straight line and charges only the residual. A monotonic drift of any size costs nothing; bumps around the line still cost, and still forget over L. Q is tridiagonal

(Appendix I), while the subtracted term has rank two and is dense, because the fitted line listens to every bin. $\mathsf{P}_{\rm mem}\mathsf{U} = 0$, $\mathsf{P}_{\rm mem}$ is positive semidefinite with rank $n-2$, and a group of one or two bins has $\mathsf{P}_{\rm mem} = 0$.

D The bias continuity equation, step by step

The matter overdensity is $\delta_m = (\rho_m - \bar\rho_m)/\bar\rho_m$, with $\bar\rho_m(t)$ the mean density; the galaxy-count overdensity $\delta_{\rm gal}$ is defined the same way. Conservation of mass and of galaxy number gives, in comoving coordinates and to linear order (the convective term $\mathbf{v}\cdot\nabla\delta$ is second order),

$$\frac{\partial\delta_m}{\partial t} \;=\; -\frac{1}{a}\,\nabla\cdot\mathbf{v}_m, \qquad \frac{\partial\delta_{\rm gal}}{\partial t} \;=\; -\frac{1}{a}\,\nabla\cdot\mathbf{v}_{\rm gal}, \tag{25}$$

with t cosmic time and $\mathbf{v}_m$, $\mathbf{v}_{\rm gal}$ the peculiar velocities. On large scales galaxies fall in the same potential as the matter and carry no pressure of their own, so $\mathbf{v}_{\rm gal} = \mathbf{v}_m$ (no velocity bias; galaxies as test particles in Tegmark and Peebles [10]). Subtracting the two equations and integrating from z_c,

$$\delta_{\rm gal}(z) - \delta_m(z) \;=\; \delta_{\rm gal}(z_c) - \delta_m(z_c)\,: \tag{26}$$

the excess clumping keeps its initial value. With $\delta_{\rm gal}(z_c) = b_c\,\delta_m(z_c)$ and $\delta_m(z) = \delta_m(z_c)\,D(z)/D(z_c)$,

$$\delta_{\rm gal}(z) \;=\; \delta_m(z)\left[\,1 + \big(b_c - 1\big)\,\frac{D(z_c)}{D(z)}\,\right], \tag{27}$$

and dividing by $\delta_m(z)$ gives Eq. (16); dividing by the growth-tied bias gives Eq. (17). As D grows the frozen term is divided by an ever larger number, so $b \to 1$: gravity debiases a conserved sample.

Tegmark and Peebles also allow stochastic bias, with b the ratio of the galaxy to the matter fluctuation amplitude and ρ_{gm} the correlation coefficient of the two fields. The same argument evolves both: with D normalized to one today and the subscript zero marking today,

$$b_0 = \left[(1-D)^2 + 2D(1-D)\,b\,\rho_{gm} + D^2 b^2\right]^{1/2}, \qquad \rho_{gm,0} = \frac{(1-D) + D\,b\,\rho_{gm}}{b_0}, \tag{28}$$

with b and ρ_{gm} the values at the epoch of growth factor D. Both approach one as clustering grows, and for $\rho_{gm} = 1$ Eq. (28) collapses to $b_0 - 1 = (b-1)\,D$, Eq. (16) again. For a DESI sample the assumption that breaks is conservation (Appendix B.3); no velocity bias and linear theory hold well on the scales kept. The script of Appendix L integrates Eq. (25) and recovers Eq. (16) to 2×10^{-9}.

E The BGS flux-limit bias model, step by step

This appendix derives the BGS drift of the baseline prior: a 22.3% span, a convex shape, and a 2.4% rms residual around the best line (the function `bgs_flux_limit_bias` of `plots/make_theory_data.py`).

E.1 The four steps

Step 1: the flux limit becomes a luminosity threshold. BGS Bright keeps every galaxy brighter than $r = 19.5$ [23] (r the magnitude band). The apparent magnitude r and the absolute magnitude M are related by $r = M + 25 + 5\log_{10}[d_L/(h^{-1}\,\mathrm{Mpc})] + K(z)$, with d_L the luminosity distance and $K(z)$ the K-correction [22]. With $K = 0$ the faintest absolute magnitude that passes at redshift z is

$$M_{\rm lim}(z) - 5\log_{10} h \;=\; 19.5 - 25 - 5\log_{10}\frac{d_L(z)}{h^{-1}\,\mathrm{Mpc}}, \qquad d_L(z) \;=\; (1+z)\,\chi(z), \tag{29}$$

with $\chi(z)$ the comoving distance of flat ΛCDM with $\Omega_m = 0.3$ and h the Hubble constant in units of 100 km/s/Mpc.

Steps 2 and 3: each bin is a threshold sample, and the threshold sets the bias. At redshift z the sample holds every galaxy with $M < M_{\rm lim}(z)$, a threshold that brightens with distance. The SDSS luminosity-threshold samples at $z < 0.25$ [20] give

$$b(>L_{\rm th}) \;=\; 1.06 + 0.21\,(L_{\rm th}/L_*)^{1.12}, \qquad L_{\rm th}/L_* \;=\; 10^{-0.4\,(M_{\rm lim}-M_*)}, \tag{30}$$

with $M_* - 5\log_{10} h = -20.44$ the characteristic magnitude of the SDSS r-band luminosity function [21]. The model applies the fit at every redshift, without evolution, and reads the published $b\,(\sigma_8/0.8)$ as b.

Step 4: the deviation from the growth-tied shape. Because B_g is free, only the shape of $b(z)$ matters; the amplitude makes the mean deviation over the group's bin centers vanish:

$$\delta(\bar z_i) \;=\; \frac{b(\bar z_i)\,D(\bar z_i)}{B} - 1, \qquad B \;=\; \frac{1}{n}\sum_{j=1}^{n} b(\bar z_j)\,D(\bar z_j). \tag{31}$$

At the six $\Delta z = 0.05$ bin centers of BGS, $z = 0.125$ to 0.375, the threshold brightens from -18.56 to -21.24, so $L_{\rm th}/L_*$ grows from 0.18 to 2.10 and the bias from 1.09 to 1.54, while D falls from 0.937 to 0.824. The deviation runs from -8.0% to $+14.4\%$, the 22.3% span of Figure 11.

E.2 Why the drift is convex

The bias rises by 41% while D falls by 12%, so the deviation inherits the shape of the bias curve, which is flat at low redshift and steep at high redshift: below L_* the term $0.21\,(L_{\rm th}/L_*)^{1.12}$ is small against 1.06 (0.03 at the first center), the threshold crosses L_* near $z = 0.27$, and beyond it the term rises exponentially (0.48 at the last center). A straight line through a convex curve leaves a U-shaped residual, $(+3.1, -0.5, -2.5, -2.6, -0.8, +3.3)\%$ here. Adding a quadratic term cuts the residual from 2.4% to 0.1% rms, so the 2.4% is the curvature of the drift, the one part a free tilt cannot absorb.

E.3 What the model leaves out

None of the omissions changes the qualitative shape, a smooth, monotonic, convex rise toward the far edge of the group.

- **Evolution of the threshold-bias relation.** The bias at fixed luminosity grows with redshift (Eq. (16)), which steepens the drift; M_* brightens by 1–2 magnitudes per unit redshift [21], which flattens it, but only by 0.3–0.6 magnitudes at the last center against the 2.68 of the threshold.

- **No K-correction.** A positive K, 0.37 magnitudes at $z = 0.4$ for a spectrum flat in wavelength and more for a red one [22], makes the true threshold brighter at high redshift and steepens the drift.

- **Only $r < 19.5$.** BGS Faint, at $19.5 < r < 20.175$ [23], lowers the threshold by 0.675 magnitudes and flattens the drift; the fiber-magnitude cut and the color selection change the galaxy mix in a way Eq. (30) cannot follow.

The model is an order-of-magnitude estimate, not a prediction of the measured DESI BGS bias. Its reliable content is the shape of the drift and its size to a factor of order unity, which is what the baseline prior uses: $\sigma_g = 3\%$ for BGS, from the 2–3% curvature.

F The accuracy test

This appendix gives the method of Section 6 and of the break test of Section 5.

F.1 The shift formula

Let m be the expected data vector (the band powers), C_d its covariance, D the matrix of derivatives of m with respect to the parameters (sans-serif, not the growth factor D), and $F = \mathsf{D}^\mathsf{T}\mathsf{C}_d^{-1}\mathsf{D} + F_{\rm priors}$ the Fisher matrix of one fit, its priors included (Appendix K). If the truth moves the data vector by δm, the maximum of the posterior moves, to first order, by

$$\delta p \;=\; F^{-1}\mathsf{D}^\mathsf{T}\mathsf{C}_d^{-1}\,\delta m\,: \tag{32}$$

the score (the gradient of the log posterior) at the old maximum becomes $\mathsf{D}^\mathsf{T}\mathsf{C}_d^{-1}\delta m$ instead of zero, and the curvature F converts it into a displacement. With C_w the fit's marginalized 2×2 covariance of (w_0, w_a) and δw the (w_0, w_a) entries of δp, the shift

$$\Delta_w \;=\; \sqrt{\delta w^\mathsf{T}\,\mathsf{C}_w^{-1}\,\delta w} \tag{33}$$

is the displacement of the dark-energy ellipse in units of its own size; Ω_m and A_s shifts are $|\delta p|/\sigma_p$. The threshold is 1/3, a shift that widens the effective error by a few percent. Ellipse and response use the same F, so a rigid model is punished twice: it absorbs little of the drift, and its ellipse is small.

A drift of the bin biases enters as $\delta m = \sum_i (\partial m/\partial\beta_i)\, B_g\, \delta_i$, with β_i the amplitudes of Eq. (11); under model A the column is that of Eq. (65) divided by $D(\bar z_i)$, so a drift of b_1 drags b_2 and b_{s^2} along. The growth-tied prior is evaluated in its own parameter basis. For the free prior the bias derivatives span the drift exactly, so the cosmology does not move, and that accuracy is bought with the information the prior gives up.

F.2 The drift laws

The conserved-population drift applies to every group: the deviation of Eq. (17), anchored at the group's mean redshift, nearly a straight line of a few percent across each group. The BGS flux-limit drift applies to BGS alone: the convex deviation of Appendix E, a 22% span whose best line leaves a 2.4% rms curvature. Both come from `plots/make_theory_data.py`; only their shapes matter, because the free amplitudes remove the level.

F.3 Random truths

A random truth sets the bias of bin i in group g to

$$b_i \;=\; \frac{B_g}{D(\bar{z}_i)}\left(1+\delta_i^{\rm drift}+\varepsilon_i\right) \tag{34}$$

at the fiducial cosmology, with $\delta_i^{\rm drift}$ the known drift (the flux limit in BGS, the conserved population elsewhere) and ε_i one draw from $\mathcal{N}(0,\Sigma)$, whose covariance $\Sigma_{ij} = \sigma_g^2 e^{-|\bar{z}_i-\bar{z}_j|/L}$ inside a group, zero across groups, is the kernel of one of the two stated priors. Two families of 2000 truths are drawn, baseline and pessimistic scatter, and every truth is fitted under every prior, each centered on the growth-tied shape. No fit is told which truth it faces.

Equation (32) is linear in δm, and δm in the bias deviations, so the (w_0, w_a) rows of the response collapse into one $2 \times n_{\rm bins}$ matrix A per prior, layout and variant,

$$\delta w \;=\; \mathsf{A}\,\delta, \qquad \delta_i \;=\; \delta_i^{\rm drift}+\varepsilon_i, \tag{35}$$

and the 2000 truths cost one matrix product. The band powers are quadratic in the bin biases, so the dropped second-order term is a few percent of the shift for 1–3% residuals and perhaps 10–20% for the 22% BGS drift. Where the fit's prior equals the truth's scatter, the average shift is not a bias: that fit's ellipse is wider than the growth-tied one precisely because it marginalizes over the scatter the truth has.

F.4 The test statistic

A fit is correctly specified when its ellipse covers the truth with the stated probability. The squared distance of the fitted (w_0, w_a), $\hat{w}$, from the truth $w_{\rm true}$ in units of the ellipse is $\chi^2_w = (\hat{w}-w_{\rm true})^{\mathsf{T}}\mathsf{C}_w^{-1}(\hat{w}-w_{\rm true})$. Averaged over the bias history and the data noise,

$$\langle\chi^2_w\rangle \;=\; \Delta^2_{\rm drift} \;+\; {\rm tr}\Big(\mathsf{C}_w^{-1}\mathsf{A}\,\Sigma\,\mathsf{A}^{\mathsf{T}}\Big) \;+\; {\rm tr}\big(\mathsf{C}_w^{-1}\mathsf{C}_w^{\rm noise}\big)\,: \tag{36}$$

the shift $\Delta_{\rm drift}$ of the known drift alone, squared; the scatter that the fit's bias parameters, held back by their prior, did not follow; and the noise. $\mathsf{C}_w^{\rm noise}$ is the share of C_w from the data noise and the Gaussian priors other than the bias prior; the cross terms average to zero. Eq. (36) is evaluated in closed form, the exact average over infinitely many universes; the 2000 draws only make the percentile bars of Figure 3, and carry no noise. The quoted value is $\eta = \sqrt{\langle\chi^2_w\rangle/2}$. An honest ellipse gives $\langle\chi^2_w\rangle = {\rm tr}(\mathsf{C}_w^{-1}\mathsf{C}_w) = 2$, one unit per parameter, and growing both axes by η restores that.

Coverage of one parameter. For one parameter p, in units of its marginalized error σ_p,

$$\langle \chi_p^2 \rangle \;=\; \mu^2 \;+\; \mathsf{A}_p^{\mathsf{T}} \Sigma\, \mathsf{A}_p \;+\; \nu, \tag{37}$$

with μ the shift of the known drift alone, A_p the row of the response that carries the bias deviations into p, divided by σ_p, and ν the noise term, the (p,p) entry of $F^{-1}F_0F^{-1}$ over σ_p^2, where $F_0 = \mathsf{D}^{\mathsf{T}}\mathsf{C}_d^{-1}\mathsf{D} + F_{\rm other}$ holds the data and the Gaussian priors other than the bias prior. The bias prior adds a positive semidefinite precision, $F \succeq F_0$, so $F^{-1}F_0F^{-1} \preceq F^{-1}$ and $\nu \le 1$ (with $X \preceq Y$ meaning $x^{\mathsf{T}}Xx \le x^{\mathsf{T}}Yx$ for every x): the rms displacement beyond the noise is at least $(\eta^2-1)^{1/2}$ error bars.

The response is linear and both the noise and the residual are Gaussian, so the pull $\zeta = (\hat{p} - p_{\rm true})/\sigma_p$ is Gaussian with mean μ and variance $V = \langle \chi_p^2 \rangle - \mu^2$, and the 1σ bar covers the truth in a fraction

$$P(|\zeta| < 1) \;=\; \Phi\!\left(\frac{1-\mu}{\sqrt{V}}\right) - \Phi\!\left(\frac{-1-\mu}{\sqrt{V}}\right) \tag{38}$$

of the universes, with $\Phi(x) = \frac{1}{2}\big[1 + \mathrm{erf}(x/\sqrt{2})\big]$. When the prior absorbs the known drift ($\mu \approx 0$),

$$P(|\zeta| < 1) \;=\; 2\,\Phi(1/\eta) - 1 : \tag{39}$$

68% at $\eta = 1$, 50% at 1.5, 38% at 2. For $\eta \ge 1$ a nonzero μ only lowers the coverage, so Eq. (39) is then an upper bound. For the baseline, pessimistic and free priors it agrees with Eq. (38) within 0.12 percentage points; for the growth-tied prior only Eq. (38) is right.

Under the pessimistic scatter and model A, on $\Delta d_{\rm com} = 62$ Mpc, $\Delta z = 0.02$, 0.05 and 0.2, the bar covers the truth in 67, 66, 60 and 68% of universes for Ω_m and 67, 66, 64 and 68% for A_s under the baseline prior; 64–68% under the pessimistic prior; 68.3% under the free prior; and 12, 1.9, 0.4 and 7.4% for Ω_m under the growth-tied prior, where Eq. (39) would give 22, 16, 15 and 23%. For (w_0, w_a) the coverage depends on how the excess of $\langle \chi_w^2 \rangle$ splits between the two directions, so only η is quoted.

The baseline prior against a pessimistic truth. The baseline prior charges a large penalty for bin biases that follow the 3% wiggles of a pessimistic truth, which change over $\Delta z \approx 0.05$, so they follow only in part and the fit moves the cosmology to absorb the rest, differently for every truth. At $\eta = 2$ the real scatter of the best-fit (w_0, w_a) is twice as wide per axis as the reported ellipse, and the FoM four times too large. The opposite mismatch gives η below 1: a conservative ellipse.

The matched case. Collect all parameters in θ, with every prior centered on the fiducial, and let $\Delta\theta$ be the truth's departure from it: zero in the cosmology and in every parameter with another Gaussian prior, $B_g\,\delta_i$ in the amplitude β_i. With $\mathbf{n}_d$ the data noise, of covariance C_d, and $F = \mathsf{D}^{\mathsf{T}}\mathsf{C}_d^{-1}\mathsf{D} + F_{\rm bias} + F_{\rm other}$, where $F_{\rm bias}$ is the bias prior's precision, the linear response gives the fit's error

$$\hat{\theta} - \theta_{\rm true} \;=\; F^{-1}\big(\mathsf{D}^{\mathsf{T}}\mathsf{C}_d^{-1}\mathbf{n}_d \;-\; F_{\rm bias}\,\Delta\theta\big) : \tag{40}$$

the noise, of covariance $F^{-1}F_0F^{-1}$, plus the part of the truth's departure that the bias prior held back. In the matched case $\mathsf{P}_{\rm mem}$ is built from the truth's kernel Σ, and every projection of that form obeys

$$\mathsf{P}_{\rm mem}\,\Sigma\,\mathsf{P}_{\rm mem} \;=\; \mathsf{P}_{\rm mem} : \tag{41}$$

with $\mathsf{Q} = \Sigma^{-1}$ and $\mathsf{M} = (\mathsf{U}^\mathsf{T}\mathsf{Q}\mathsf{U})^{-1}$, $\Sigma\,\mathsf{P}_{\rm mem} = \mathsf{I} - \mathsf{U}\mathsf{M}\mathsf{U}^\mathsf{T}\mathsf{Q}$, and $\mathsf{P}_{\rm mem}\mathsf{U} = 0$ removes the second term. The scatter part of Eq. (40) then has covariance $F^{-1}F_{\rm bias}F^{-1}$, the two covariances add up to F^{-1}, and the scatter and noise terms of Eq. (36) sum to exactly 2: the widening the prior adds to the ellipse equals the error the scatter inflicts. The matched value is $\eta = (1+\Delta^2_{\rm drift}/2)^{1/2}$ for (w_0, w_a) and $(1+\Delta^2_{{\rm drift},p})^{1/2}$ for one parameter; under model A it stays within 6% of 1 for (w_0, w_a), and a single parameter reaches 1.11 (Ω_m on $\Delta z = 0.05$, where the BGS flux limit is absorbed least well).

When a fit is conservative. Write $\mathsf{P}_{\rm mem}$ in its null-space form, Eq. (61), with the fit's kernel $\Sigma_{\rm fit}$, and let $\Sigma_{\rm true}$ be the truth's. If $\mathsf{N}^\mathsf{T}\Sigma_{\rm true}\mathsf{N} \preceq \mathsf{N}^\mathsf{T}\Sigma_{\rm fit}\mathsf{N}$, then $\mathsf{P}_{\rm mem}\Sigma_{\rm true}\mathsf{P}_{\rm mem} \preceq \mathsf{P}_{\rm mem}$ and $\eta \le (1+\Delta^2_{\rm drift}/2)^{1/2}$: a prior wider than the truth wastes information but does not understate the error. The free prior has $F_{\rm bias} = 0$, so only the noise remains in Eq. (40) and $\eta = 1$ for every truth expressible through the bin biases.

F.5 The same test for Ω_m and A_s

Ω_m and A_s are scored the same way, each in units of its own marginalized error, with $\eta = \langle\chi^2_p\rangle^{1/2}$, the case $n_{\rm dof} = 1$ of Eq. (8). They can react more strongly to a bias error than (w_0, w_a), because a bias error moves the clustering amplitude $b_i^2 A_s D^2(\bar z_i)$ of bin i, which A_s scales directly and Ω_m moves through the growth factor.

G The projected precision for two bins

Everything essential already happens with two bins, of amplitudes β_1 and β_2, common fiducial B, spread σ_β and correlation r:

$$\mathsf{C} \;=\; \sigma_\beta^2 \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}, \qquad \mathsf{C}^{-1} \;=\; \frac{1}{\sigma_\beta^2\,(1-r^2)} \begin{pmatrix} 1 & -r \\ -r & 1 \end{pmatrix}. \tag{42}$$

The penalty of a displacement $\Delta\beta = (\beta_1 - B,\ \beta_2 - B)^\mathsf{T}$ is $\Delta\beta^\mathsf{T}\,\mathsf{C}^{-1}\Delta\beta$. A uniform shift, $\Delta\beta = c\,(1,1)^\mathsf{T}$, costs

$$\frac{2c^2}{\sigma_\beta^2\,(1+r)} \;\neq\; 0, \tag{43}$$

but it is a change of B_g, which the growth-tied model leaves free. A shape change, $\Delta\beta = d\,(1,-1)^\mathsf{T}$, costs

$$\frac{2d^2}{\sigma_\beta^2\,(1-r)}, \tag{44}$$

which grows without bound as $r \to 1$. At $r = 0.670$ (Appendix A.2) the shape costs $6.1\,d^2/\sigma_\beta^2$ and the uniform shift $1.2\,c^2/\sigma_\beta^2$. Both rows of C^{-1} have the same sum, so $\mathsf{C}^{-1}\mathbf{1} = [\sigma_\beta^2(1+r)]^{-1}(1,1)^\mathsf{T}$ and Eq. (15) gives

$$\mathsf{P} \;=\; \frac{1}{2\sigma_\beta^2(1-r)} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad \Delta\beta^\mathsf{T}\mathsf{P}\,\Delta\beta \;=\; \frac{(\beta_1-\beta_2)^2}{2\sigma_\beta^2(1-r)}. \tag{45}$$

The denominator is the prior variance of the jump, $2\sigma_\beta^2(1-r)$: *the prior constrains the jump, not the level.*

H The projection as a marginalization

Equation (15) removes the prior's information about one direction, the uniform shift: the projected penalty of $\Delta\beta$ is the plain penalty of what remains after subtracting the best uniform shift, the generalized-least-squares mean $\hat{c} = \mathbf{1}^\mathsf{T}\mathsf{C}^{-1}\Delta\beta/\mathbf{1}^\mathsf{T}\mathsf{C}^{-1}\mathbf{1}$. The derivation below shows that this is the correct prior, not just one possible repair.

H.1 Marginalizing a flat-prior amplitude

Model the n amplitudes of a group as

$$\beta \;=\; B\,\mathbf{1} + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \mathsf{C}), \tag{46}$$

with a flat prior on the scalar B: the model of Eq. (9), linearized around the fiducial. Integrating B out,

$$p(\beta) \;\propto\; \int dB\, \exp\Big[-\tfrac{1}{2}\big(\beta - B\,\mathbf{1}\big)^\mathsf{T}\mathsf{C}^{-1}\big(\beta - B\,\mathbf{1}\big)\Big]. \tag{47}$$

With $S = \mathbf{1}^\mathsf{T}\mathsf{C}^{-1}\mathbf{1}$, completing the square gives

$$\big(\beta - B\mathbf{1}\big)^\mathsf{T}\mathsf{C}^{-1}\big(\beta - B\mathbf{1}\big) \;=\; S\,\big(B - B_*\big)^2 \;+\; \beta^\mathsf{T}\mathsf{C}^{-1}\beta \;-\; \frac{\big(\mathbf{1}^\mathsf{T}\mathsf{C}^{-1}\beta\big)^2}{S}, \qquad B_* = \frac{\mathbf{1}^\mathsf{T}\mathsf{C}^{-1}\beta}{S}. \tag{48}$$

The B integral contributes only the constant $\sqrt{2\pi/S}$, and what survives is

$$p(\beta) \;\propto\; \exp\Big[-\tfrac{1}{2}\,\beta^\mathsf{T}\mathsf{P}\,\beta\Big], \qquad \mathsf{P} \;=\; \mathsf{C}^{-1} - \frac{\mathsf{C}^{-1}\mathbf{1}\,\mathbf{1}^\mathsf{T}\mathsf{C}^{-1}}{\mathbf{1}^\mathsf{T}\mathsf{C}^{-1}\mathbf{1}}, \tag{49}$$

exactly Eq. (15). In matrix language, the exponent of Eq. (47) is a Gaussian in (β, B) with precision

$$\mathsf{F}_{(\beta,B)} \;=\; \begin{pmatrix} \mathsf{C}^{-1} & -\,\mathsf{C}^{-1}\mathbf{1} \\ -\,\mathbf{1}^\mathsf{T}\mathsf{C}^{-1} & \mathbf{1}^\mathsf{T}\mathsf{C}^{-1}\mathbf{1} \end{pmatrix}, \tag{50}$$

and removing B leaves the Schur complement $\mathsf{P} = \mathsf{F}_{\beta\beta} - \mathsf{F}_{\beta B}\,\mathsf{F}_{BB}^{-1}\,\mathsf{F}_{B\beta}$: the block corrected downward by what B could absorb. P is the statement "*B_g is sampled with a flat prior and marginalized*", folded into one matrix.

I Many bins: the tridiagonal inverse

I.1 Three bins: total correlation versus direct links

With the covariance $\mathsf{C}_{ij} = \sigma_\beta^2 \exp(-|\bar{z}_i - \bar{z}_j|/L)$ of Eq. (12), positive definite for distinct redshifts, three evenly spaced bins with neighbor correlation r give

$$\mathsf{C} \;=\; \sigma_\beta^2 \begin{pmatrix} 1 & r & r^2 \\ r & 1 & r \\ r^2 & r & 1 \end{pmatrix}, \qquad \mathsf{C}^{-1} \;=\; \frac{1}{\sigma_\beta^2\,(1 - r^2)} \begin{pmatrix} 1 & -r & 0 \\ -r & 1 + r^2 & -r \\ 0 & -r & 1 \end{pmatrix}. \tag{51}$$

Bins 1 and 3 *are* correlated (r^2 in C), yet their entry in C^{-1} is zero: C gives the total correlation, what δ_1 says about δ_3 when nothing else is known, while C^{-1} gives the direct one, what δ_1 still

says once δ_2 is known, which is nothing. The exponential kernel makes this exact because its correlations multiply along the chain,

$$e^{-(\bar{z}_3-\bar{z}_1)/L} \;=\; e^{-(\bar{z}_3-\bar{z}_2)/L}\, e^{-(\bar{z}_2-\bar{z}_1)/L}; \tag{52}$$

a Gaussian kernel $e^{-\Delta z^2/L^2}$ does not factorize, and its inverse is dense.

I.2 The chain of links

Because correlations multiply along the chain, the joint density of the deviations factorizes into neighbor conditionals,

$$\begin{aligned} p(\delta_1,\dots,\delta_n) \;&=\; p(\delta_1)\, p(\delta_2 \mid \delta_1)\cdots p(\delta_n \mid \delta_{n-1}), \\ \big\langle \delta_{k+1} \mid \delta_k \big\rangle = r_k\, \delta_k, \qquad &\mathrm{Var}\big(\delta_{k+1} \mid \delta_k\big) = \sigma_g^2\,\big(1-r_k^2\big), \end{aligned} \tag{53}$$

with $r_k = e^{-(\bar{z}_{k+1}-\bar{z}_k)/L}$. Splitting $\delta_{k+1} = \lambda\,\delta_k + e$ with e uncorrelated with δ_k gives $\lambda = r_k$ and $\mathrm{Var}(e) = \sigma_g^2(1-r_k^2)$, and earlier bins add nothing: for $j \le k$,

$$\mathrm{Cov}(e,\delta_j) \;=\; \sigma_g^2\, e^{-(\bar{z}_k-\bar{z}_j)/L}\left[e^{-(\bar{z}_{k+1}-\bar{z}_k)/L} - r_k\right] \;=\; 0. \tag{54}$$

Taking $-2\ln p$ gives the penalty as a sum over neighbor links,

$$\delta^{\mathsf{T}}\big(\sigma_g^2\mathsf{R}\big)^{-1}\delta \;=\; \frac{\delta_1^2}{\sigma_g^2} \;+\; \sum_{k=1}^{n-1} \frac{\big(\delta_{k+1} - r_k\,\delta_k\big)^2}{\sigma_g^2\,\big(1-r_k^2\big)}\,: \tag{55}$$

directly only neighbors talk; in total the influence propagates down the chain (the Markov property).

I.3 The tridiagonal inverse

Collecting the coefficient of every product $\delta_i\delta_j$ in Eq. (55) reads off the precision matrix. For the amplitudes ($\Delta\beta = B_g\,\delta$, $\sigma_\beta = \sigma_g B_g$) it is tridiagonal,

$$\sigma_\beta^2\,\big(\mathsf{C}^{-1}\big)_{kk} = \begin{cases} \dfrac{1}{1-r_1^2}, & k=1, \\[2mm] \dfrac{1}{1-r_{k-1}^2} + \dfrac{r_k^2}{1-r_k^2}, & 1<k<n, \\[2mm] \dfrac{1}{1-r_{n-1}^2}, & k=n, \end{cases} \qquad \sigma_\beta^2\,\big(\mathsf{C}^{-1}\big)_{k,k+1} = \frac{-\,r_k}{1-r_k^2}\,: \tag{56}$$

the off-diagonal entry is the link of pair k, and the diagonal entry of a bin collects the links touching it. For even spacing the interior diagonal becomes $(1+r^2)/(1-r^2)$ and

$$\mathsf{C}^{-1} \;=\; \frac{1}{\sigma_\beta^2\,(1-r^2)} \begin{pmatrix} 1 & -r & & & \\ -r & 1+r^2 & -r & & \\ & -r & \ddots & \ddots & \\ & & \ddots & 1+r^2 & -r \\ & & & -r & 1 \end{pmatrix}, \tag{57}$$

which for $n=3$ is Eq. (51).

I.4 The projection, group by group, and its limits

Each group receives the projection of Eq. (15) built from its own C, and the total prior precision is block diagonal. The kernel model contains both extremes. As $L \to \infty$, every $r_k \to 1$, the jump variance of every neighbor pair vanishes, and only the uniform shift is left: the growth-tied model, whatever σ_g. As $\sigma_g \to \infty$ at fixed L, $\mathsf{P} \to 0$: the free model.

J The memory projection, step by step

A group has n bins, C is the kernel covariance of the residuals, $\mathsf{Q} = \mathsf{C}^{-1}$, $\mathsf{U} = [\,\mathbf{1},\ v\,]$ the $n \times 2$ design matrix of Eq. (23) with v the centered redshifts, and $\theta = (c, s_g)^{\mathsf{T}}$ a line's level and slope. (For the bin amplitudes every covariance carries a factor B_g^2 that cancels in each identity below.)

J.1 The line fit, the split, and the marginalization

Minimizing $(\delta - \mathsf{U}\theta)^{\mathsf{T}}\mathsf{Q}\,(\delta - \mathsf{U}\theta)$ over the line gives the normal equations

$$\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\,\hat{\theta} \;=\; \mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\delta, \qquad \hat{\theta} = (\hat{c}, \hat{s}_g)^{\mathsf{T}}, \tag{58}$$

with $\mathsf{U}^{\mathsf{T}}\mathsf{Q}\mathsf{U}$ positive definite whenever the group has two distinct redshifts. Writing $\delta = \mathsf{U}\hat{\theta} + \delta_\perp$, the residual obeys $\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\delta_\perp = 0$, the cross terms vanish, and the penalty splits,

$$\delta^{\mathsf{T}}\mathsf{Q}\,\delta \;=\; \hat{\theta}^{\mathsf{T}}\big(\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\big)\hat{\theta} \;+\; \delta_\perp^{\mathsf{T}}\mathsf{Q}\,\delta_\perp, \qquad \delta^{\mathsf{T}}\mathsf{P}_{\mathrm{mem}}\,\delta \;=\; \delta_\perp^{\mathsf{T}}\mathsf{Q}\,\delta_\perp : \tag{59}$$

the projection keeps the residual's penalty and discards the line's. A pure line has $\delta_\perp = 0$, so $\mathsf{P}_{\mathrm{mem}}\mathsf{U} = 0$; the residuals obey two linear constraints, so the rank is $n - 2$; and $\delta_\perp^{\mathsf{T}}\mathsf{Q}\,\delta_\perp \geq 0$ makes it positive semidefinite.

Marginalization gives the same matrix. With $\delta \sim \mathcal{N}(\mathsf{U}\theta, \mathsf{C})$ and $\theta \sim \mathcal{N}(0, \Lambda)$, $\Lambda = \mathrm{diag}(W^2, \tau^2)$ the prior variances of the level and the slope, the marginal covariance $\mathsf{C} + \mathsf{U}\Lambda\mathsf{U}^{\mathsf{T}}$ inverts by the Woodbury identity,

$$\big(\mathsf{C} + \mathsf{U}\Lambda\mathsf{U}^{\mathsf{T}}\big)^{-1} \;=\; \mathsf{Q} - \mathsf{Q}\,\mathsf{U}\big(\Lambda^{-1} + \mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\big)^{-1}\mathsf{U}^{\mathsf{T}}\mathsf{Q}, \tag{60}$$

which becomes $\mathsf{P}_{\mathrm{mem}}$ as $\Lambda^{-1} \to 0$, with a leading correction falling as $1/W^2$.

J.2 The null-space form

Let the columns of the $n \times (n-2)$ matrix N span the combinations of bins that no straight line can produce, $\mathsf{N}^{\mathsf{T}}\mathsf{U} = 0$. Then

$$\mathsf{Q} - \mathsf{Q}\,\mathsf{U}\big(\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\big)^{-1}\mathsf{U}^{\mathsf{T}}\mathsf{Q} \;=\; \mathsf{N}\big(\mathsf{N}^{\mathsf{T}}\mathsf{C}\,\mathsf{N}\big)^{-1}\mathsf{N}^{\mathsf{T}}. \tag{61}$$

Both sides annihilate U and map CN to N, and the columns of CN and U together span the whole space (if $\mathsf{CN}x + \mathsf{U}y = 0$, multiplying by N^{T} gives $x = 0$, then $y = 0$), so the two matrices are equal. The prior reads off the combinations $\mathsf{N}^{\mathsf{T}}\delta$ that no line can make and charges each by the inverse of its own prior covariance. With the single column $\mathsf{U} = \mathbf{1}$ the same identity gives Eq. (15).

J.3 Worked examples

For $n = 2$, U is square and invertible and the projection vanishes; for $n = 1$, N has no columns. For three evenly spaced bins with spacing Δ and $r = e^{-\Delta/L}$, $v = (-\Delta, 0, \Delta)^{\mathsf{T}}$ and the only direction orthogonal to $\mathbf{1}$ and v is the discrete curvature $\mathsf{N} = (1, -2, 1)^{\mathsf{T}}$, with prior variance $\mathsf{N}^{\mathsf{T}}\mathsf{C}\,\mathsf{N} = 2\sigma_g^2(1-r)(3-r)$, so

$$\delta^{\mathsf{T}}\mathsf{P}_{\mathrm{mem}}\,\delta \;=\; \frac{\big(\delta_1 - 2\delta_2 + \delta_3\big)^2}{2\,\sigma_g^2\,(1-r)(3-r)}\,: \tag{62}$$

the memory model charges only the curvature, where the kernel model of Eq. (51) taxes a drift as heavily as a zigzag of the same step.

J.4 The universal-kriging predictor

Predict bin $k+1$ from the first $k \geq 2$ bins, with flat priors on θ. With C_k the past kernel block, U_k the past rows of U, $u_{k+1} = (1, v_{k+1})^{\mathsf{T}}$ and γ the kernel covariances between bin $k+1$ and the past,

$$\mathrm{E}\big[\delta_{k+1} \mid \delta_{1..k}, \theta\big] \;=\; u_{k+1}^{\mathsf{T}}\theta \;+\; \gamma^{\mathsf{T}}\mathsf{C}_k^{-1}\big(\delta_{1..k} - \mathsf{U}_k\,\theta\big). \tag{63}$$

With a flat prior, θ is replaced by the fit $\hat{\theta}$ of Eq. (58) on the past block, and by Eq. (52) $\gamma^{\mathsf{T}}\mathsf{C}_k^{-1} = r_k\,e_k^{\mathsf{T}}$, with e_k the k-th unit vector. Substituting gives Eq. (22). With a Gaussian slope of variance τ^2, dropping the kernel term far beyond L gives the correlation of Eq. (21).

K How the prior enters the Fisher matrix

A Gaussian prior adds its precision to the Fisher matrix,

$$F \;=\; F_{\mathrm{data}} \;+\; F_{\mathrm{priors}}, \tag{64}$$

with one block per prior. $\mathsf{P}_{\mathrm{mem}}$ of each group is added to the rows and columns of that group's amplitudes β_i for the baseline and pessimistic priors; the growth-tied prior instead sums each group's amplitude columns into one (Appendix M.4). A bin that lost all its band powers to the scale cut has zero rows in F_{data}: under the free prior its amplitude is dropped, while a correlated prior ties it to its neighbors and keeps the matrix invertible as long as the group's data fix its level and tilt.

L Numerical verification

The script `verify_prior.py`, in this folder, checks every identity of these appendices with the cocoa Python: 23 checks, each printing one PASS or FAIL line with the measured deviation. They cover the two-bin, three-bin, even and uneven tridiagonal inverses; the projection's null direction, semidefiniteness and rank; the Pythagorean split and the Schur complement; the neighbor-prediction leftover and, by Monte Carlo, the jump variance with the worked example of Appendix A.2; the continuity equation against Eq. (16), the conserved deviations and the neighbor rms values; and for the memory model the annihilation of U with rank $n-2$, the $1/W^2$ Woodbury limit, the null-space identity, the three-bin curvature penalty, the vanishing

for one and two bins, the kriging predictor, the end-to-end correlation, the straight-line fits of Appendix C.4 and the LRG3 magnitude-limit rate. The theory figures are drawn from data written by `plots/make_theory_data.py`, with the same growth factor and fiducial amplitudes.

M Implementation and the Fisher construction

M.1 Models A and B in the likelihood

The project likelihood (`likelihood/desi_cl.py`) has two bias options, `higher_order_bias`: 0 is linear bias, and 1 the one-loop model with every term per bin ($b_{1,i}$, Δ_2, Δ_s, α_0, α_2, SN_0, SN_2; $b_3 = 0$), in which a zero amplitude switches a term off at no cost. Models A and B are two parameter files of option 1, which state the grouping (`scripts/make_desi_params.py --model-a` or `--model-b`); the example yamls `EXAMPLE_EVALUATE_MODEL_A1`–`A4` and `B1`–`B4` run them on the four layouts, in layout order. The likelihood computes $b_{2,i}$ and $b_{s^2,i}$ from $b_{1,i}$ with Eq. (5) and passes them to CosmoLike per bin.

- **Counterterms.** CosmoLike's k^2 slot is $2\,b_1 b_K\,k^2 P_{\mathrm{mm}}$, with b_K its per-bin derivative-bias coefficient and k in units of H_0/c, so the likelihood sets $b_K = \alpha_0/[2\,b_1(c/H_0)^2]$, with $c/H_0 = 2997.92\,h^{-1}\mathrm{Mpc}$. α_0 therefore multiplies the nonlinear matter spectrum, where DESI's definition multiplies the linear one (`notes/higher_order_bias`, subsection "One likelihood option, every term per bin"). α_2, in $h^{-2}\,\mathrm{Mpc}^2$, has its own setter and multiplies the linear spectrum, as in Eq. (6).
- **Stochastic terms.** In Eq. (7), $P_{ii} = 1/\bar{n}_i$ is fixed by the bin's surface density, so that the projection with the raw distribution gives the angular shot noise. A cross pair of two bins of one tracer sample takes the mean of their two powers; a pair of two samples takes none, because different galaxies share no stochastic term.
- **Projection.** The low-rank non-Limber engine projects each bin's full one-loop $P_{gg,i}$, counterterm included, and the adjacent cross pairs carry the bilinear one-loop pair spectrum (`notes/higher_order_bias`, subsection "Separability and the D^4 rule"). Appendix P shows why the counterterm requires this engine.

M.2 The derivatives

The derivatives of the data vector m (the band powers) come in three kinds. *Stencil derivatives*, from a five-point stencil of the model A likelihood: the seven cosmological parameters ($10^9 A_s$, the spectral index n_s, the Hubble constant H_0, the baryon density Ω_b, Ω_m, w_0, w_a), the six group amplitudes B_g and the velocity dispersions. *Per-bin operator columns* $\partial m/\partial b_{1,i}$, $\partial m/\partial b_{2,i}$, $\partial m/\partial b_{s^2,i}$, $\partial m/\partial \alpha_{0,i}$ and, for model B, $\partial m/\partial \alpha_{2,i}$, each at fixed values of the others, from central differences through the likelihood's setters (steps of 1% of b_1, 0.02 in b_2 and b_{s^2}, $5\,h^{-2}\,\mathrm{Mpc}^2$ in α_0). The bins shifted together form one color of a greedy coloring of the cross-pair graph, in which two bins joined by a measured cross pair never share a color, so each spectrum involves at most one shifted bin; with four colors per layout each operator costs eight evaluations instead of two per bin. *Stochastic columns*: the exact templates of Eq. (7).

Model A ties b_2 and b_{s^2} to b_1, so the derivative with respect to a bin's linear bias is the chain-rule sum

$$\left.\frac{dm}{db_{1,i}}\right|_{\mathrm{A}} = \frac{\partial m}{\partial b_{1,i}} + b_2^{\mathrm{L}\prime}(b_{1,i})\,\frac{\partial m}{\partial b_{2,i}} - \frac{4}{7}\,\frac{\partial m}{\partial b_{s^2,i}}, \tag{65}$$

with $b_2^{\mathrm{L}\prime}$ the slope of the Lazeyras fit. Summed over a group with weights $1/D(\bar{z}_i)$, these columns must reproduce the stencil derivative of the amplitude B_g (Appendix O).

M.3 The variants and the Fisher matrices

The parameters are the cosmology and the velocity dispersions (with the Gaussian priors of the linear-bias run), the variant's higher-order parameters, and one amplitude $\beta_i = b_{1,i}\, D(\bar{z}_i)$ per bin (Eq. (11)), whose column is Eq. (65) divided by $D(\bar{z}_i)$. The higher-order columns of group g are sums over its bins,

$$\frac{\partial m}{\partial \Delta_{2,g}} = \sum_{i \in g} \frac{\partial m}{\partial b_{2,i}}, \qquad \frac{\partial m}{\partial t_{2,g}} = \sum_{i \in g} (\bar{z}_i - \bar{z}_g)\, \frac{\partial m}{\partial b_{2,i}}, \tag{66}$$

and likewise for the other terms, with $t_{2,g}$ the tilt of Δ_2 in group g. Models A and B + tilts add the tilt columns, with flat priors, in every group of more than one bin: the free straight line in redshift that s_g gives the linear bias; the exact-calibration configuration of Appendix Q has no higher-order column.

The data Fisher matrix is $F_{\mathrm{data}} = \mathsf{D}^{\mathsf{T}} \mathsf{C}_d^{-1} \mathsf{D}$ on the kept entries, with D the matrix of the columns and C_d the covariance of Section 2; the Gaussian priors add to its diagonal, and the linear-bias prior adds its projected precision to the amplitude block (Appendix K). In the $k_{\max}$ scan each variant's derivatives are a fixed combination $\mathsf{D} = \mathsf{B}\mathsf{M}$ of the base columns B, so $F_{\mathrm{data}} = \mathsf{M}^{\mathsf{T}} (\mathsf{B}^{\mathsf{T}} \mathsf{C}_d^{-1} \mathsf{B})\, \mathsf{M}$ on the entries each cut keeps: 336 systems (four layouts, seven cuts, four variants, three priors), 324 with a fit and 12 without any band. Every Fisher matrix is inverted through a pseudo-inverse (Appendix O).

M.4 The priors, the response and the truths

- **Linear-bias priors.** Growth-tied sums the amplitude columns of each group into one; the baseline and pessimistic priors add the projected precision of Eq. (23) to the amplitude block; free adds nothing.
- **Known drifts and random truths.** The two drift laws of Appendix F.2 and the 2000 draws per scatter family of the linear-bias run, with its seed and order; a drift δ_i of the linear bias gives $\delta m = \sum_i (\partial m / \partial \beta_i)_{\mathrm{A}}\, \beta_i\, \delta_i$, with the column of Eq. (65), and the fit moves by Eq. (32).
- **Calibration drift.** One deterministic truth that moves only the offsets, $\delta m = \sum_i [(\partial m / \partial b_{2,i})\, \Delta_{2,i} + (\partial m / \partial b_{s^2,i})\, \Delta_{s,i}]$, with $\Delta_{2,i} = (\bar{z}_i - \bar{z}_g)/(\bar{z}_{\max} - \bar{z}_{\min})$ and $\Delta_{s,i} = \Delta_{2,i}/2$, where $\bar{z}_{\max}$ and $\bar{z}_{\min}$ are the largest and smallest mean redshifts of the group's bins. It changes by 1 in Δ_2 and 0.5 in Δ_s across the group and has zero mean over its bins; a one-bin group gets zero.
- **Break truths.** δm is the α_2 or SN_2 column times the truth's value, or, for the offsets, the b_2 and b_{s^2} columns times $\pm(1, 0.5)$ for model A and $\pm(2, 1)$ for model B, summed over every bin, or the calibration drift above. The random α_2 truths draw one value per group from model B's prior, with the same 2000 standard normal vectors at every cut.

Each figure has its own folder under `plots/`, with its data (`data.json`), the script that draws it from that file alone, and the PDF. The data come from the notebooks `EXAMPLE_EVALUATE_FISHER_MODEL_A.ipynb` and `EXAMPLE_EVALUATE_FISHER_MODEL_B.ipynb`.

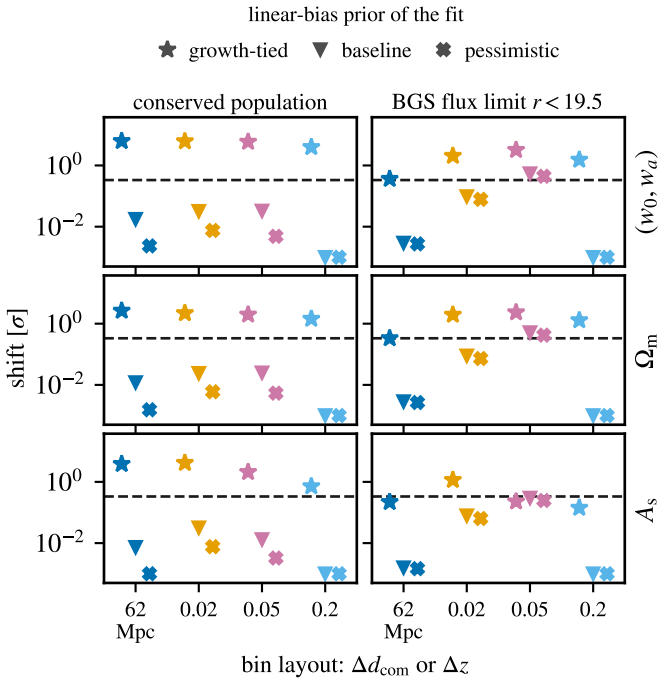


Figure 12: The shift of a model A fit under each known drift alone, the conserved population (left) and the BGS $r < 19.5$ flux limit (right): (w_0, w_a) in units of the fit's ellipse, Ω_m and A_s in their own σ. Markers: the linear-bias prior; colors: the layouts; dashed: $\sigma/3$. Points on the bottom axis: values below 10^{-3}. Omitted: the free prior, which absorbs both drifts exactly. $k_{\max} = 0.075\,h/\mathrm{Mpc}$ quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

N The known drifts

Figure 12 draws the shift of each known drift alone under model A. The free prior absorbs both drifts exactly. The growth-tied prior stays excluded: the conserved-population drift moves its (w_0, w_a) by several σ, and still by 1.5–2.6σ under model A + tilts. The baseline and pessimistic priors absorb both drifts below $\sigma/3$ in every score but the BGS flux limit on $\Delta z = 0.05$, a layout that resolves BGS into only six bins. There both priors land above $\sigma/3$ in (w_0, w_a) and in Ω_m (0.54 and 0.51 for the baseline prior, 0.44 and 0.42 for the pessimistic); this residual drift lifts the pessimistic prior's η on that layout to its maximum of 1.08 (Appendix F.4). Under model A + tilts both priors stay below $\sigma/10$.

O Numerical checks

- **The chain rule.** For every group, the per-bin columns of Eq. (65), summed with the weights $1/D(\bar{z}_i)$, reproduce the stencil derivative of the group amplitude B_g to 1.4×10^{-9}, 1.5×10^{-9}, 1.2×10^{-9} and 1.3×10^{-9} of the stencil column's largest entry (layout order).
- **The stochastic templates.** The order-0 templates of the SN_2 run equal the cached SN_0 templates exactly, and its fiducial vector has $\Delta\chi^2 = 0$ against the derivative cache.

SN_2 is linear to 3×10^{-14}.

- **The α_2 columns.** The forward and backward differences agree to 1.6×10^{-10} of each column, and the vector after the steps equals the one before bit for bit. The model A FoMs of the $k_{\max}$ scan equal those of the model A notebook at $0.075\,h/\mathrm{Mpc}$ to 6.5×10^{-10}.

- **The matched-prior identity.** When a fit's prior equals the truth's scatter, the identity $\mathsf{P\Sigma P} = \mathsf{P}$ of Appendix F.4 makes the scatter and noise terms of $\langle \chi^2_w \rangle$ sum to exactly 2: $\max |\langle \chi^2_w \rangle - \Delta^2_{\mathrm{drift}} - 2| = 1.7 \times 10^{-9}$ over every layout and variant. The random α_2 truths with α_2 removed give $\eta = 1$ to 1.4×10^{-9}.

- **Exact recovery.** The free prior must absorb any truth of the linear bias; its largest mean (w_0, w_a) shift over the random truths is 1.4×10^{-11}.

- **Directions without information.** On $\Delta d_{\mathrm{com}} = 62$ Mpc at $k_{\max} = 0.075\,h/\mathrm{Mpc}$ only 7 of the 18 BGS bins keep any band, and under model A + tilts on the free prior the flat-prior BGS tilts leave one combination of parameters with no information, along which a plain inverse amplified round-off into a spurious 2σ shift. The notebooks therefore scale every Fisher matrix to unit diagonal, diagonalize it, and drop the directions below 10^{-12} of the largest eigenvalue: one of the 48 systems at this cut and 20 of the 336 of the scan, with largest components on (w_0, w_a), Ω_m and A_s of 1.3×10^{-11} and 7.2×10^{-10}, pure nuisance combinations.

- **The α_0 step.** At $\alpha_0 = -5\,h^{-2}\,\mathrm{Mpc}^2$ the factor $b_1^2 + \alpha_0 k^2$ turns negative above $k \approx 0.6\,h/\mathrm{Mpc}$, where the low-rank engine floors the spectrum it factorizes; on $\Delta z = 0.05$, a step of 5 against 1 changes the α_0 columns by at most 3.5×10^{-4} of their largest entry, and a recomputation of the derivatives is bitwise identical to the cache.

- **The loop templates.** With $b_3 = 0$, CosmoLike's C-FAST-PT loop templates and Python FAST-PT agree on the $\Delta d_{\mathrm{com}} = 62$ Mpc vector, pairs included, to $\Delta\chi^2 \leq 3.5 \times 10^{-5}$ (`notes/higher_order_bias`, section "Validation").

P The counterterm requires the low-rank engine

CosmoLike has two non-Limber engines for C_ℓ^{gg}. FKEM, the FFTLog method of Fang et al. [8], projects the linear term exactly and moves everything nonlinear, the counterterm included, through its Limber difference. The low-rank engine factorizes each bin's full one-loop P_{gg}, less the white-noise constant of the loops (which Limber projects exactly), and projects it without the Limber approximation.

On $\Delta z = 0.05$ the low-rank response $\partial C_\ell / \partial \alpha_0$ is 7.3 times the Limber one at $\ell = 20$, 1.8 times at $\ell = 50$, 1.26 times at $\ell = 100$ and about 1.05 times at $\ell = 200$–400 (median over the kept autos). Limber evaluates the power at the transverse wavenumber $k = (\ell + 1/2)/\chi$ only, while a thin bin with sharp edges also passes radial modes of higher k, which the k^2 weight amplifies. The low-rank response is converged: raising `nonlimber_accuracyboost` from 32 to 128, the rank from 5 to 10, or the overall accuracy boost from 1 to 2 changes it by a $\Delta\chi^2$ far below one. For a one-sigma α_0 the two engines' responses differ by tens of units of χ^2.

Q Comparison with linear bias and with exact calibrations

Neither comparison is a result of this note: the linear-bias run uses a model that `notes/linear_bias_kmax` shows is biased at $k_{\rm max} = 0.075\,h/\mathrm{Mpc}$, and exact calibrations assume that b_2 and b_{s^2} are known exactly. Both sets of FoMs are in `plots/baseline_ladder/data.json`.

Linear bias. Model A keeps about 78, 73, 72 and 50% of the linear-bias run's FoM(w_0, w_a) on the baseline prior, and model A + tilts about 72, 58, 52 and 30% on the pessimistic prior (layout order). The loss grows as the bins widen, likely because a wide-layout group holds few bins, so one offset per group is close to one per bin and competes with each bin's linear bias.

Exact calibrations. With the calibrations exact and $\alpha_0 = \mathrm{SN}_0 = \mathrm{SN}_2 = 0$, the FoM exceeds that of the linear-bias run under every prior and on every layout, on the baseline prior about 1.6 times on $\Delta d_{\rm com} = 62$ Mpc and 2.5 times on $\Delta z = 0.05$. The one-loop correction is then a known function of $b_{1,i}$ and of the matter power spectrum; the loops grow as the fourth power of the growth factor and the linear term as b_1^2 times its square, so their ratio carries information on the amplitude of the matter fluctuations. That information is only as good as the calibrations: the calibration drift then shifts (w_0, w_a) by 1.0–2.0σ under the growth-tied prior and 0.07–0.46σ under the other three. The offsets, counterterm and stochastic terms of model A remove it, and model A sits below the linear-bias run under every prior and on every layout.

R Limitations

- **Linear response.** The shifts and $\sqrt{\langle\chi^2\rangle/n_{\rm dof}}$ are first order in the truth's departure from the fiducial. The dropped second-order term is a few percent of the shift for 1–3% residuals and perhaps 10–20% for the 22% BGS drift (Appendix F.3).
- **Theory truths, not mocks.** Residuals rougher than the pessimistic prior, and calibrations that drift nonlinearly within a group, are untested.

S Open issues

- **One-loop terms of the covariance in Limber.** The covariance carries the one-loop terms of model A through the Limber approximation, and its low-ℓ non-Limber recompute is linear. In the lowest bands of thin bins that leaves an error of at most 1.3% of C_ℓ.
- **The scale cut was not rebuilt on the one-loop model.** On $\Delta z = 0.05$ at $k_{\rm max} = 0.075\,h/\mathrm{Mpc}$ the UV fraction of Eq. (1), applied to the α_0 template $\partial C_\ell/\partial\alpha_0$, is a median 0.076 and a maximum 0.14 on the kept bands, against 0.017 and 0.052 for the fiducial one-loop vector, and about half of each bin's SN_2 auto kernel comes from line-of-sight wavenumbers above the bin's own cut: part of what the forecast learns about α_0 and SN_2 comes from scales the cut is meant to exclude.

- **Negative α_0 in chains.** At $\alpha_0 = -88.9\,h^{-2}\,\mathrm{Mpc}^2$, one prior width below zero, the spectrum the low-rank engine factorizes, P_{gg} less its white-noise constant, turns negative above $k \approx 0.14\,h/\mathrm{Mpc}$; the engine floors 47% of its grid, and the vector curves in α_0. The Fisher forecast at $\alpha_0 = 0$ does not see this, but a chain that explores negative α_0 does. Neither remedy, a loop taper or a positivity prior on P_{gg} (`notes/higher_order_bias`, subsection "The white-noise limit of the renormalized loops"), is implemented.

- **The b_3 template is wrong.** CosmoLike's third-order bias template (`tab_d1d3` of C-FAST-PT) is a fixed, cosmology-independent table that disagrees with Python FAST-PT, so both models set $b_3 = 0$, as DESI's baseline does.

- **No redshift-space one-loop terms.** Both models add real-space loops to linear Kaiser distortions; model B relies on the sampled σ_v and on α_2 for the rest.

- **Adjacent cross spectra see the shared edge.** The cross spectra of adjacent thin bins are sensitive to small-scale radial physics at their common edge, as SN_2 shows; models A and B carry SN_2 for that reason, and any other radial physics on that scale reaches the cross spectra the same way.

References

[1] E. Saraivanov, H. Rubira, V. Miranda and T. Eifler, "Towards the Two-Loop EFTofLSS in Galaxy Lensing Surveys", 2026, arXiv:2603.13031.

[2] J. DeRose and S. Chen, "The Lensing Counter Narrative: An Effective Description of Small-Scale Clustering in Weak Lensing Power Spectra", 2025, arXiv:2510.18981.

[3] T. Lazeyras, C. Wagner, T. Baldauf and F. Schmidt, "Precision measurement of the local bias of dark matter halos", 2016, arXiv:1511.01096.

[4] S. Saito, T. Baldauf, Z. Vlah, U. Seljak, T. Okumura and P. McDonald, "Understanding higher-order nonlocal halo bias at large scales by combining the power spectrum with the bispectrum", 2014, arXiv:1405.1447.

[5] DESI Collaboration, "DESI 2024 V: Full-shape galaxy clustering from galaxies and quasars", 2024, arXiv:2411.12021.

[6] M. Maus, S. Chen, M. White et al., "An analysis of parameter compression and full-modeling techniques with Velocileptors for DESI 2024 and beyond", 2024, arXiv:2404.07312.

[7] The LSST Dark Energy Science Collaboration, "The LSST Dark Energy Science Collaboration (DESC) Science Requirements Document", 2018, arXiv:1809.01669.

[8] X. Fang, E. Krause, T. Eifler and N. MacCrann, "Beyond Limber: Efficient computation of angular power spectra for galaxy clustering and weak lensing", 2020, JCAP, 05, 010, arXiv:1911.11947.

[9] J. N. Fry, "The Evolution of Bias", 1996, ApJ, 461, L65.

[10] M. Tegmark and P. J. E. Peebles, "The Time Evolution of Bias", 1998, ApJ, 500, L79, arXiv:astro-ph/9804067.

[11] D. A. Wake et al., "The 2dF-SDSS LRG and QSO Survey: evolution of the clustering of luminous red galaxies since $z = 0.6$", 2008, MNRAS, 387, 1045, arXiv:0802.4288.

[12] R. Tojeiro et al., "The progenitors of present-day massive red galaxies up to $z \sim 0.7$: finding passive galaxies using SDSS-I/II and SDSS-III", 2012, MNRAS, 424, 136, arXiv:1202.6241.

[13] H. Guo et al., "The clustering of galaxies in the SDSS-III Baryon Oscillation Spectroscopic Survey: luminosity and color dependence and redshift evolution", 2013, ApJ, 767, 122, arXiv:1212.1211.

[14] R. Zhou et al., "The clustering of DESI-like luminous red galaxies using photometric redshifts", 2021, MNRAS, 501, 3309, arXiv:2001.06018.

[15] R. Zhou et al., "Target Selection and Validation of DESI Luminous Red Galaxies", 2023, AJ, 165, 58, arXiv:2208.08515.

[16] R. Zhou et al., "DESI luminous red galaxy samples for cross-correlations", 2023, JCAP, 11, 097, arXiv:2309.06443.

[17] M. White et al., "Cosmological constraints from the tomographic cross-correlation of DESI luminous red galaxies and Planck CMB lensing", 2022, JCAP, 02, 007, arXiv:2111.09898.

[18] N. Sailer et al., "Cosmological constraints from the cross-correlation of DESI luminous red galaxies with CMB lensing from Planck PR4 and ACT DR6", 2025, JCAP, 06, 008, arXiv:2407.04607.

[19] N. Sailer et al., "Evolution of structure growth during dark energy domination: insights from the cross-correlation of DESI galaxies with CMB lensing and galaxy magnification", 2025, Phys. Rev. D, arXiv:2503.24385.

[20] I. Zehavi et al., "Galaxy Clustering in the Completed SDSS Redshift Survey: The Dependence on Color and Luminosity", 2011, ApJ, 736, 59, arXiv:1005.2413.

[21] M. R. Blanton et al., "The Galaxy Luminosity Function and Luminosity Density at Redshift $z = 0.1$", 2003, ApJ, 592, 819, arXiv:astro-ph/0210215.

[22] M. R. Blanton and S. Roweis, "K-corrections and Filter Transformations in the Ultraviolet, Optical, and Near-Infrared", 2007, AJ, 133, 734, arXiv:astro-ph/0606170.

[23] C. Hahn et al., "The DESI Bright Galaxy Survey: Final Target Selection, Design, and Validation", 2023, AJ, 165, 253, arXiv:2208.08512.

[24] A. Raichoor et al., "Target Selection and Validation of DESI Emission Line Galaxies", 2023, AJ, 165, 126, arXiv:2208.08513.

[25] A. Rocher et al., "The DESI One-Percent Survey: exploring the halo occupation distribution of emission line galaxies with AbacusSummit simulations", 2023, JCAP, 10, 016, arXiv:2306.06319.